

BCLC strategy for prognosis prediction and treatment recommendations: The 2026 update

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Summary

The 2026 BCLC update incorporates recent therapeutic advances in HCC into its recommendations, while preserving simplicity. Each disease stage remains directly linked to its evidence-based first-line treatment option. The BCLC algorithm provides guidance for evaluating patient prognosis and proposes a therapeutic strategy according to the current scientific evidence. Tumour boards require structured methodologies to address clinical complexity. Beyond published evidence, decision-making should incorporate biological, psychosocial, and contextual factors that may affect morbidity, feasibility, and patient vulnerability. Such a multidimensional approach ensures treatments remain evidence-based and patient-centred. The updated clinical decision-making chapter embeds the CUSE (Complexity, Uncertainty, Subjectivity, Emotion) framework. CUSE turns unavoidable doubt into a shared, iterative process to: (i) define the therapeutic goal (survival, tumour control, quality of life, etc.); (ii) grade each option, noting evidence strength and gaps; (iii) align choices with comorbidities, feasibility, oncologic risk, and patient values and goals; and (iv) select a plan with regular check-ins as new information or needs arise.

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Introduction

Across successive updates,^{1–8} the Barcelona Clínic Liver Cancer (BCLC) prognosis and staging system has refined hepatocellular carcinoma (HCC) staging and treatment allocation. We highlight the developments that justify a strategic shift and explain why other promising, but still nascent, approaches are not yet ready for inclusion. All findings are graded by study design – recognizing both the insights and limitations of observational data – and, as before, we engage external experts to broaden the BCLC group's perspective. Here, we have incorporated new datasets analysable by BCLC stage, and we have omitted data without clear, stage-specific results. The 2025 update also preserves all previous prognostic refinements – such as albumin–bilirubin (ALBI) and alpha-fetoprotein (AFP) stratification; a detailed assessment of decompensating events (such as prior variceal bleeding and ascites grade or response); performance status (ECOG-PS); and frailty, which is defined as a multidimensional syndrome of reduced reserve and increased vulnerability. It further provides a clearer definition of multifocal disease in intermediate stage B. Finally, we have introduced the CUSE (Complexity,

Uncertainty, Subjectivity, Emotion) framework to structure person-centred decision making when evidence about separate alternatives is non-comparable or lacking (Fig. 1A,B).

The supplementary material is structured in three parts: SUP-A – literature interpretation and tailored approaches for BCLC-0/A and BCLC-B; SUP-B – tailored approaches for BCLC-C and BCLC-D; SUP-C – non-proven options/sequences and patient experience/quality of life.

Prognosis prediction and patient characterisation

Life expectancy in HCC is modelled by three interacting forces: tumour burden, liver function, and cancer-related symptoms. Hence, a single multivariable model to predict outcomes across all evolutionary stages of the disease would bear excessive complexity.⁹ In addition, some parameters that would be key for some stages would be of no value in others, as happens at the time of treatment recommendation. Therefore, from its inception, the BCLC strategy has followed a two-step approach: (i) delineate broad evolutionary stages with the major prognostic variables, and (ii) develop more granular

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Keypoints

- The 2026 BCLC strategy refines clinical decision-making and treatment recommendations in line with recent scientific advances.
- Prognosis in HCC depends on the dominant drivers – tumour burden and/or liver function – while management adapts to evolving evidence.
- The 2026 update preserves the 2022 staging system but further specifies BCLC-B, detailing the term “multifocal” as: >3 nodules, or ≤3 nodules with at least one >3 cm.
- First recommended treatment options for each stage are listed preferentially according to level of scientific evidence for overall survival benefit.
- Personalized treatment recommendations are guided by the CUSE framework, which integrates Complexity, Uncertainty, Subjectivity, and Emotion with the patient's full clinical, biological, and psychosocial profile.
- CUSE is a flexible, human-centred decision-support framework – not a rigid protocol – that helps make uncertainty explicit and fosters shared, multidisciplinary, and evidence-based decision-making across diverse clinical settings.

prognostic and therapeutic models within each stage. This architecture was first advanced in 2000¹ and has been validated repeatedly and adopted by most guidelines worldwide as the starting point for describing individuals with HCC. It has also served to characterize the study population in most HCC-related investigations, including phase III clinical trials.

Tumour burden should be assessed with contrast-enhanced multi-phase computed tomography (CT) or magnetic resonance imaging (MRI) by liver cancer-dedicated radiologists to avoid under- or over-staging, as even one additional nodule or a more accurate diameter may shift stage. Chest CT is advised to complete staging when appropriate.^{10,11} Contrast-enhanced ultrasound is accepted as a complementary tool for diagnosis when CT/MRI are inconclusive,¹⁰ while positron emission tomography is not recommended.¹²

Since 2018,⁵ liver function has not been stratified solely by the Child–Pugh (CP) score, because CP class A may still include patients with decompensation, such as ascites, variceal bleeding, or prior hepatic encephalopathy, which are clear signs of severe portal hypertension and reflect advanced liver disease. The incorporation of the ALBI score refines mortality risk prediction across all BCLC stages.^{13,14} For decompensated disease, sequential refinements of the model for end-stage liver disease score improve short-term prognostication.^{15–17}

Cirrhosis status represents the first branching point in the framework (Fig. 2). If cirrhosis is present, prognosis depends on whether it is compensated or decompensated, independent of the CP score. In compensated cirrhosis, outcomes are mainly driven by the biology and extent of HCC. In decompensated cirrhosis, residual liver function – alongside HCC extent and tumour biology – takes precedence. Grading ascites refines the prognosis prediction as subclinical ascites detected at imaging is not equivalent to tense ascites.¹⁸ Some individuals develop HCC without cirrhosis, or after sustained virologic response to HCV regardless of fibrosis regression. In these cases, prognosis depends almost entirely on HCC biology.

Importantly, decompensation does not represent an absolute and irreversible contraindication to specific therapies. Patients who achieve recompensation¹⁹ after treating the aetiological factor of the underlying liver disease or an adequate controlled state through best supportive measures for cirrhosis

complications (e.g. diuretics, dietary interventions, or treatment of encephalopathy, variceal bleeding, or renal dysfunction) may re-enter specific therapeutic pathways, as reflected in the CUSE framework (Fig. 2).

Cancer-related symptoms – captured objectively by the (ECOG-PS) – remain a robust, stage-independent predictor of overall survival (OS) and are a common stratification factor in phase III trials of systemic therapy.²⁰ By design, (ECOG-PS) reflect cancer-attributable symptoms rather than those due to comorbidities or underlying liver disease. However, attribution can be challenging in decompensated cirrhosis and even in compensated cirrhosis with marked fatigue. ECOG-PS is a useful tool to estimate functional status, but in advanced cirrhosis or elderly patients its interpretation is complex. Reduced activity or fatigue may not always be due to HCC, but rather cirrhosis-related complications (ascites, encephalopathy, sarcopenia) or frailty associated with aging and comorbidities. In this context, complementing ECOG-PS with dedicated frailty assessment tools may better disentangle tumour-driven symptoms from other contributors, while clinical judgment remains essential to interpret patient-reported limitations.

An elevated AFP concentration signals worse prognosis and may exclude transplantation,^{21,22} but it does not preclude other therapies. Many additional biomarkers – lectin-bound AFP, Prothrombin Induced by Vitamin K Absence II (PIVKA-II), inflammatory cytokines, C-reactive protein – have been proposed, yet none has gained sufficient validation to replace the established predictors (SUP-A&B).^{23,24}

In practice, HCC staging starts with clinical assessment and imaging; prognosis is refined by liver-function tests, ECOG-PS, and biomarkers. Decisions rest with a multidisciplinary tumour board (MTB) – diagnostic and interventional radiology, hepatology, medical and radiation oncology, surgery, pathology, nuclear medicine, molecular biology, specialized nursing, palliative care, dietitian and social work – with immune-related adverse event (irAE) experts included following the introduction of immunotherapy.²⁵ This team maps stage to predict survival and selects therapy by integrating clinical, imaging, pathology, laboratory, and psychosocial data. Leadership within the MTB should be assumed by the professional best

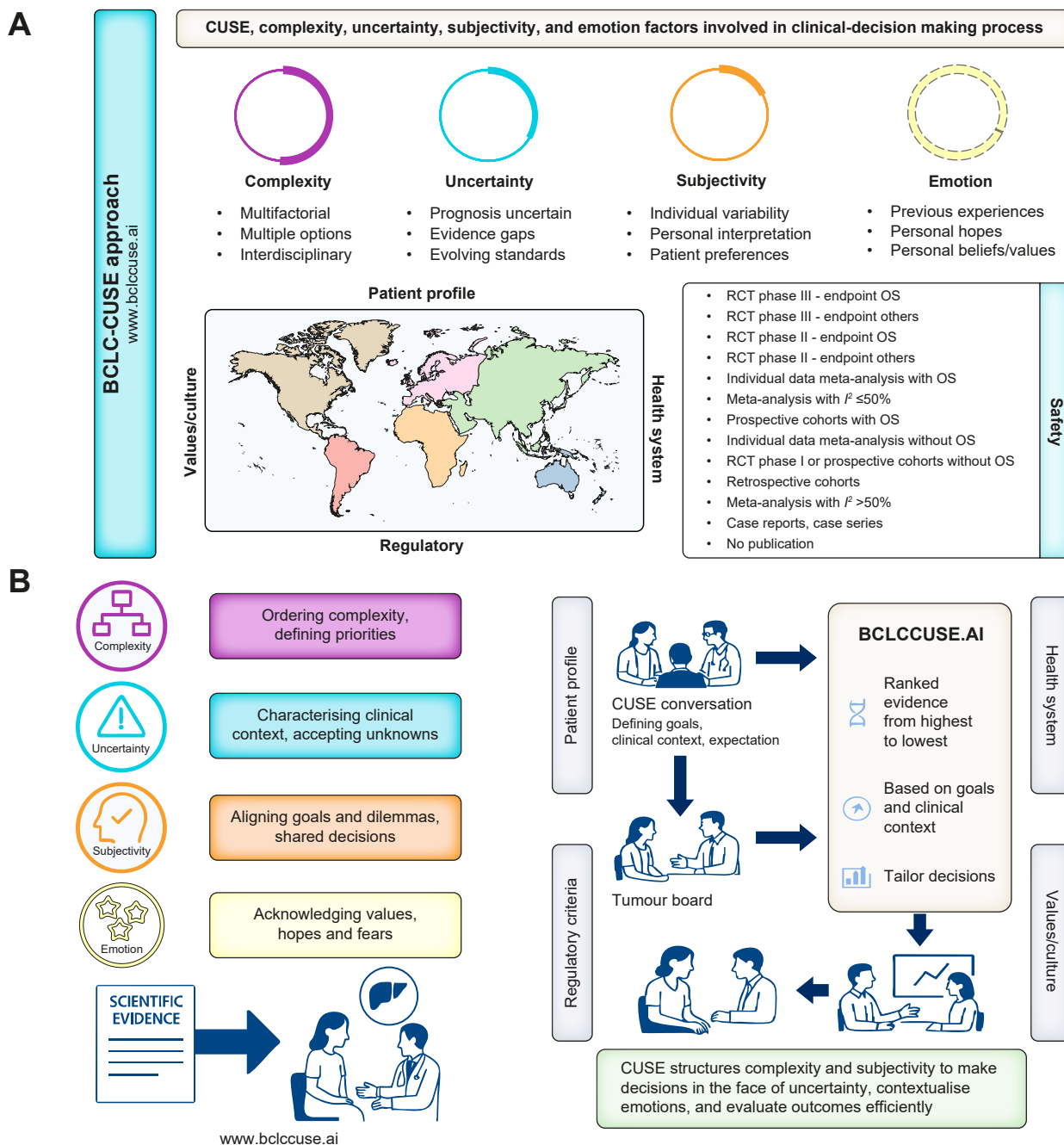


Fig. 1. CUSE framework. (A) CUSE approach: framing optimal, person-centred decisions amid uncertainty. The CUSE framework supports person-centred decision-making in HCC by integrating four dimensions – Complexity (multifactorial disease and treatment options), Uncertainty (ambiguous prognosis and evolving evidence), Subjectivity (individual variability and preferences), and Emotion (previous experiences, hopes, and personal beliefs/values) – with contextual factors such as values, regulations, and health-system capacity. The world map illustrates regional differences in contextual factors that shape the implementation of CUSE globally. Evidence tiers (right panel) span from no evidence to phase III trials with OS, ensuring safety at every level. CUSE advocates defining choices as ‘optimal’ rather than ‘best’ when evidence is lacking – avoiding false certainty – and promotes explicit acceptance of uncertainty through time-bound, reviewable decisions with clear objectives. This ensures that each person’s unique profile guides therapy beyond trial data alone. (B) CUSE framework process: The CUSE process (co-designed) provides a structured pathway for clinical decision-making in HCC. It involves: (1) establishing the goals of care; (2) assessing the clinical and social context, including hepatic reserve, performance status, frailty, comorbidities, deliverability, affordability, and caregiver capacity; (3) evaluating the evidence by level of evidence and applicability, translating population data into person-level choices, and communicating options in plain language using absolute numbers; (4) deliberating within the multidisciplinary team; (5) making a shared decision documented with rationale and consent; and (6) iterating the plan to reflect changing goals, clinical status, and evolving evidence, incorporating interim decisions in ambiguous scenarios (e.g. time-limited therapy used in downstaging) and predefined review points.

positioned to integrate scientific knowledge with organisational skills, ensuring coordinated and efficient decision-making. This approach reflects the BCLC's emphasis on humanistic management, an approach consistent with the principles of *common-sense management* described by Xavier Marcet, prioritising shared responsibility, transparent communication, and patient-centred care.

Treatment

In the following paragraphs we outline the proposed treatment options for each BCLC stage. After stage characterization, the 2026 BCLC again lists preferred options for each stage. **Treatment-stage migration (TSM)** tempers the linear algorithm (Fig. 3), permitting access to alternative treatments for earlier stages or even to liver transplantation (LT) once listing criteria are met with recognition of the increased risk of morbidity and/or the potential negative impact on expected survival. **Untreatable progression (UTP)** – first defined for transarterial chemoembolisation (TACE)²⁶ but now applied across all stages – signals failure of the current strategy and triggers transition to the treatment recommended for the next BCLC stage, even in the absence of radiological progression.

Major response after treatment – still lacking a consensus definition – remains a major source of uncertainty in HCC decision-making, as the necessity and efficacy of subsequent therapies in this setting are unknown.²⁷

Although prognosis in BCLC is mainly determined by liver function and tumour burden, additional individual factors may also influence outcomes. The BCLC 2026 model makes these determinants explicit so that they can be systematically weighed together with stage and therapy. Importantly, the BCLC system was originally conceived as a starting point at

diagnosis or initiation of therapy. Over time, concepts such as TSM and BCLC upon progression²⁸ were incorporated to capture patient trajectories. The advent of immunotherapy now raises a new challenge: defining BCLC upon response.

This notion, commonly referred to as therapeutic conversion, recognises that profound tumour regression does not mark a return to a prior disease stage, but the emergence of a distinct clinical state: a form of “treatable tumour regression” requiring its own staging framework.

Clinical decision-making

The 2022 BCLC strategy⁸ introduced a clinical decision framework, which the 2026 revision refines by integrating the CUSE approach. CUSE is a structured framework applied when there is a gap between available evidence and its application to an individual patient. It integrates clinical data, patient and treatment goals, and contextual factors to organise complexity, align dilemmas with therapeutic priorities, and apply the best available evidence. This process supports multidisciplinary teams in reaching explicit, reproducible, and patient-centred decisions.

BCLCCUSE.AI is under development and it will become accessible online. It will offer a tailored, case-specific appraisal of the evidence to reduce tumour board workload and standardise CUSE implementation across diverse settings (SUP-A). This framework clarifies the concepts and parameters that the MTB must combine into a personalised treatment plan for HCC. When interpreting evidence (SUP-A), greater weight is given to prospective studies – particularly randomised controlled trials (RCT) – while retrospective studies remain more prone to bias, even with adjustments such as propensity score matching or IPTW (inverse probability of treatment

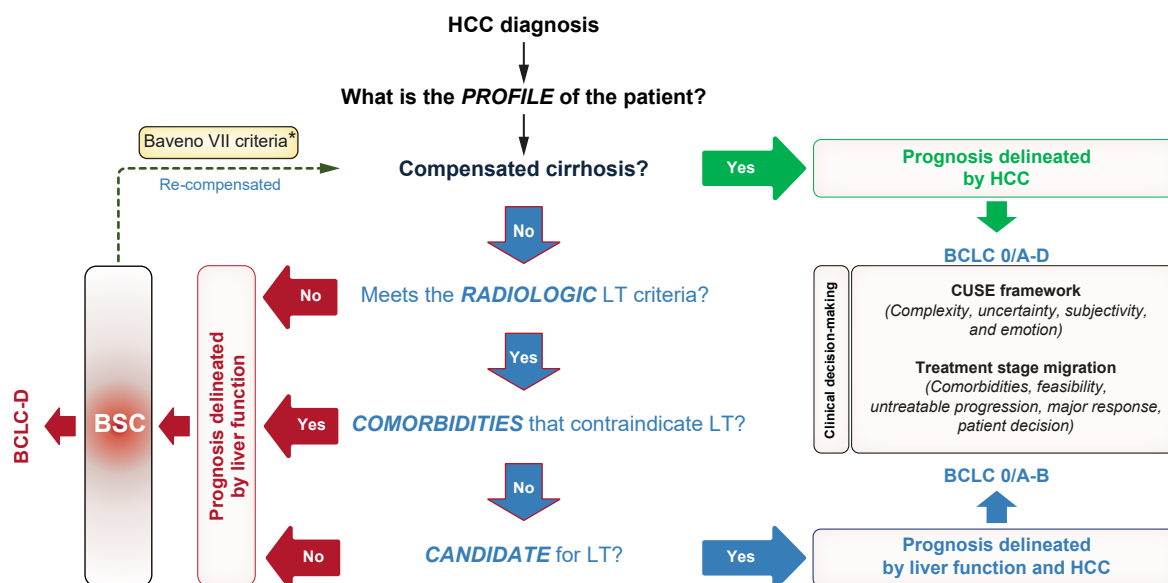


Fig. 2. Accessing the BCLC treatment flowchart. A stepwise framework for staging and prognostication in people with HCC. First, each person is assessed for compensated cirrhosis; if cirrhosis is compensated, prognosis is guided primarily by tumour biology and stage, and the person is classified as BCLC stage 0/A–D. If cirrhosis is decompensated, prognosis reflects both residual liver function and tumour burden. People who meet radiologic LT criteria and have no comorbid contraindications are considered for LT (BCLC 0/A–B). Patients with decompensated cirrhosis who do not qualify for LT, have contraindications to LT, or do not achieve recompensation are referred for best supportive care (BCLC-D). In all cases, significant cancer-related symptoms may also place a person in BCLC-D, regardless of tumour burden. The dotted arrow indicates that recompensation may allow re-entry into specific treatment pathways, in line with the CUSE framework. BCLC, Barcelona Clinic Liver Cancer; BSC, best supportive care; CUSE, Complexity, Uncertainty, Subjectivity, Emotion; HCC, hepatocellular carcinoma; LT, liver transplantation.

* Note that the concept of recompensation in patients with HCC still needs to be refined, as specific criteria have yet to be established.

weighting).^{29–31} Treatment allocation in such studies often reflects centre-specific strategies, with surgery prioritised as first-line and ablation or locoregional therapy reserved for patients unsuitable for resection. This structural selection bias cannot be fully eliminated and is explicitly acknowledged within the CUSE framework.^{29–31}

Person-oriented CUSE decision framework (Fig. 1)

CUSE addresses two recurrent evidence gaps: (i) no demonstrated OS benefit and (ii) robust OS data that are not comparable across options. It draws on Mishel's taxonomy of illness uncertainty,³² McCormick's concept of temporality,³³ and Griffiths' emphasis on exposing doubt,³⁴ to explicitly incorporate CUSE into decision-making. CUSE advocates defining choices as 'optimal' rather than 'best' when evidence is lacking, avoiding false certainty, and promotes explicit acceptance of uncertainty through time-bound, reviewable decisions with clear objectives. This ensures that each person's unique profile guides therapy beyond trial data alone.

Process (co-designed; Fig. 1B): The process involves (1) establishing the goals of care; (2) assessing the clinical and social context, including hepatic reserve, performance status, frailty, comorbidities, deliverability, affordability, and caregiver capacity; (3) evaluating the evidence by level of certainty and applicability, translating population data into person-level choices, and communicating options in plain language with absolute numbers; (4) deliberating within the multidisciplinary team; (5) making a shared decision documented with rationale and consent; and (6) iterating the plan to reflect changing goals, clinical status, and evolving evidence, with interim decisions in ambiguous scenarios (e.g., time-limited therapy, as used in downstaging) and predefined review points.

Purpose & scope: align treatment with the patient's goals, values, and life context,^{35–38} while staying evidence-anchored and safety-first. This includes clarifying goals of care (cure, survival, symptom relief, function), acceptance of risk/toxicity, day-to-day realities (work, caregiving, travel, costs), and preferred involvement in decisions.

Dimensions include: **complexity** (multifactorial comorbidities and competing treatments with equal or absent evidence); **uncertainty** (individual prognostic ambiguity, evidence gaps, rapidly evolving standards); **subjectivity** (variability in how individuals and clinicians interpret information and express preferences); and **emotion** (the psychological impact of stress, prior experiences, hopes, beliefs, or values on patients, caregivers, and clinicians).

Decision thresholds (with patient agreement): **GO** – clear goal, robust safety evidence, favourable benefit–risk, adequate hepatic reserve and performance status, and assured deliverability; **CONSIDER** – marginal eligibility or modifiable risks needing optimisation (including prehabilitation strategies to enhance physical reserve),³⁹ information gaps to address, or suitability for an approved clinical trial; **NO-GO** – high decompensation risk, absolute contraindications, uncontrolled comorbidity, non-deliverable care (logistics, access, or affordability constraints), or absence of robust safety evidence; such options are not offered outside approved trials, and care pivots to best supportive care or stage-appropriate alternatives.

When to invoke CUSE: CUSE is typically invoked when therapeutic ambiguity arises as a result of the following scenarios.

Multiple RCTs reporting improved OS

When more than one RCT demonstrates a survival benefit, but no head-to-head comparisons are available. Important considerations regarding the interpretation of median survival times are provided in **SUP-A**. Exploratory methods such as restricted mean survival time, milestone survival rates, progression-free survival (PFS), landmarks, time to treatment failure or duration of response, quality-adjusted survival, post-progression survival, or the cumulative incidence of clinically relevant events (e.g. liver failure, transplantation, variceal bleeding) may provide complementary insights. While network meta-analyses may provide additional insights in the absence of head-to-head comparisons, their interpretation requires caution. Differences in trial populations, study design, and the temporal context in which trials were conducted may introduce heterogeneity and bias, potentially limiting the validity of indirect comparisons. Therefore, although such analyses can support clinicians when carefully performed, they should not be taken as evidence of clear superiority of one systemic option over another. In line with this principle, the BCLC update has refrained from presenting cross-trial hazard ratio (HR)s or survival curve metrics, so as not to imply a hierarchy unsupported by direct evidence.

The BCLC 2026 framework considers these approaches potentially useful for providing descriptive information and generating hypotheses, particularly to anticipate expected outcomes under a given treatment. Their main value lies in informing shared decision-making between healthcare providers and patients by clarifying potential trajectories of benefit, toxicity, or quality of life, thereby guiding treatment choices in contexts of uncertainty.

RCT but with immature OS

Immature OS data – due to short follow-up and few events, especially late deaths – make interpretation challenging, particularly for curative-intent strategies (e.g. neoadjuvant or combination therapy in early to intermediate HCC). Teams must balance the statistical power of an interim readout from a phase III RCT against the smaller yet fully mature signal from a phase II study. Consistent with this principle, BCLC 2026 recommends waiting for the final phase III results and, in the meantime – while no OS benefit is proven – basing treatment recommendations on mature phase II data (safety, response, and survival), prospective projects, and individual-patient meta-analyses that provide granular OS estimates in well-characterised cohorts. When event numbers are low, stating that a therapy shows “no detrimental effect on OS” is clinically meaningless: the absence of observed harm does not imply benefit – or even neutrality.

Favourable PFS in the absence of valid OS data

PFS is informative if the trigger for it is fully defined – tumour growth, liver decompensation, death, or a composite – and the outcome for each event is clearly described, with competing risks in people with cirrhosis reported separately (Fig. 4). Development of liver decompensation that prevents treatment

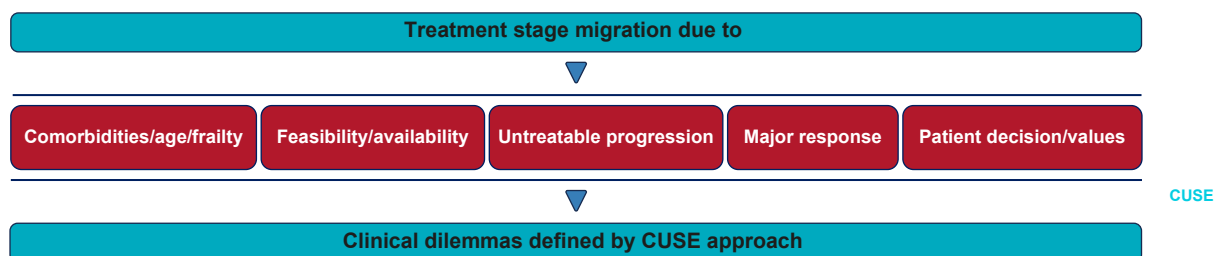


Fig. 3. Treatment-stage migration and CUSE-defined clinical dilemmas. Treatment-stage migration in HCC occurs when patients move between therapeutic categories due to factors such as comorbidities or age, feasibility constraints, untreatable tumour progression, major treatment response, or patient decisions and values. These shifts give rise to clinical dilemmas that the CUSE framework seeks to define and address. Comorbidities/age, competing health conditions or advanced age; feasibility, logistical or anatomical factors limiting intervention; untreatable progression, disease advancement beyond current therapy's scope; major response, substantial tumour regression prompting reconsideration of stage-appropriate care; patient decision/values, individual preferences and life goals guiding treatment choice. CUSE, Complexity, Uncertainty, Subjectivity, Emotion; HCC, hepatocellular carcinoma.

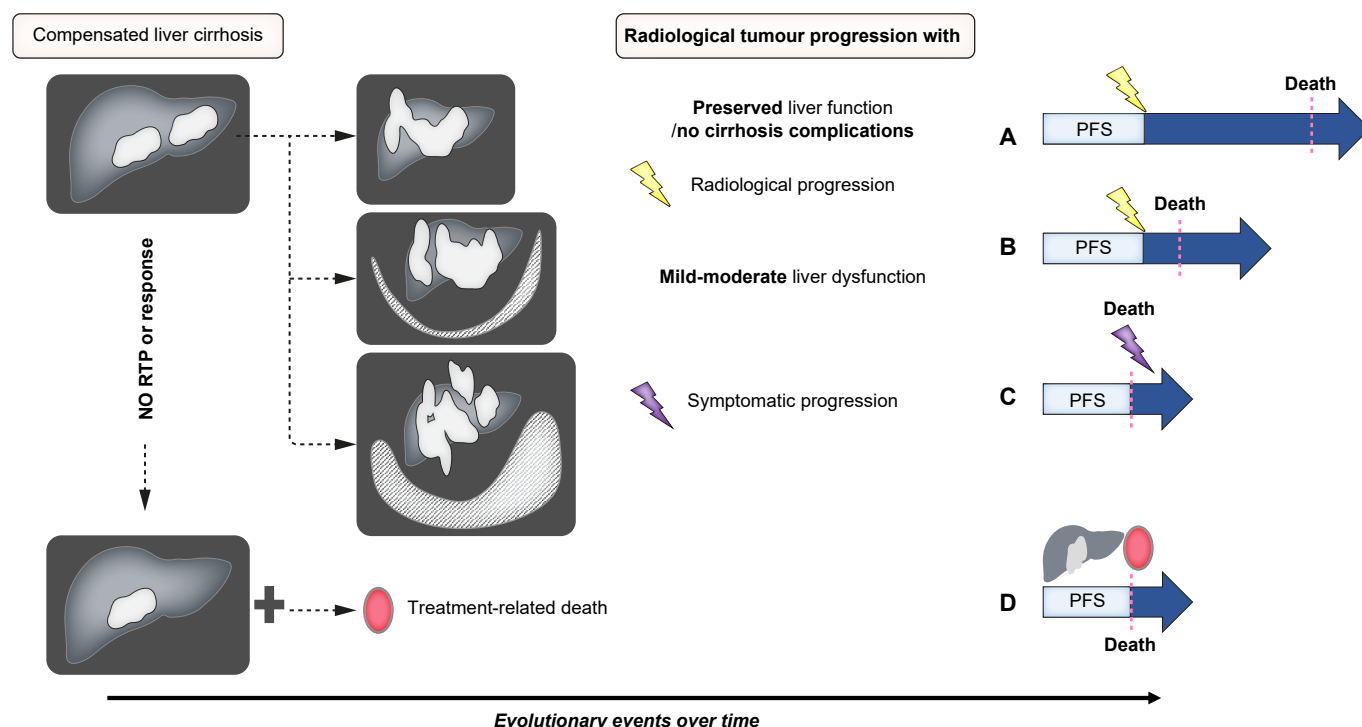


Fig. 4. Limitations of PFS as a surrogate for OS. The same PFS interval can herald very different outcomes depending on its defining event. (A) PFS also ends with progression, but of an indolent pattern that does not impact long-term survival (OS remains long). (B) PFS ends with disease progression that directly drives mortality, resulting in short OS. (C) Progression reflects aggressive tumour biology and is followed by rapid decline in OS. (D) PFS ends with treatment-related toxicity and death, completely independent of tumour behaviour, underscoring how event type affects both patient experience and the ultimate prognosis. The hatched area in B represents perilesional oedema or ascites, depending on the context, and not cirrhosis complications *per se*. OS, overall survival; PFS, progression-free survival; RTP, radiologic tumour progression.

maintenance is a clinically relevant event that may result in the 'failure of treatment strategy' endpoint as introduced in the ABC-HCC trial comparing systemic therapy vs. chemoembolisation (NCT04803994). In clinical trials, its underlying cause should be described in detail. The clinical team should accept that a PFS gain may never translate into an OS advantage, but it can justify a *temporary* decision to attempt to delay tumour progression, during which the person may reach the next-line therapy – ideally one with proven curative potential or stronger survival data. The benefit, however, remains provisional and should be framed within the inherent uncertainty. PFS remains a vulnerable

endpoint and is not considered a robust surrogate for OS in major guidelines for intermediate to advanced stage HCC.^{10,40}

Absence of RCT

When RCTs are lacking, interpretive flexibility depends on the type of available evidence (e.g. phase I–II non-randomised studies, individual patient data meta-analyses, real-world evidence, aggregated-data meta-analyses, case reports) and on the intended goal (safety or efficacy). Their main primary role is to support clinical judgment and shared decision-making by

contextualising expected benefits and risks in situations of evidence scarcity. In this context, prospective and retrospective evidence must be clearly distinguished when informing treatment strategies. Prospective single-arm phase I/II trials may justify clinical adoption when they represent the highest or most applicable evidence available, as seen with highly selective molecular therapies in other cancers. By contrast, single retrospective studies or aggregate meta-analyses of retrospective data are more prone to bias and should be regarded as hypothesis-generating unless externally validated. In addition, the growing role of real-world evidence requires careful appraisal, and its quality should be assessed using established frameworks such as the ESMO-GROW guidelines before being used to inform therapeutic recommendations.⁴¹

Expected survival after first treatment option based on evidence-based OS

Under the BCLC staging system, the expected median OS after the first therapy is >5 years for individuals at very-early and early stages (BCLC 0/A), more than 2.5 years for those at the intermediate stage (BCLC B), roughly 2 years for the advanced stage (BCLC C), and <1 year for end-stage disease (BCLC D). These estimates represent best-case outcomes with effective first-line therapy, but individual prognosis varies widely.

When sequential treatment is deployed – moving promptly to the next, stage-appropriate option once treatable progression emerges – survival can extend far beyond these medians. However, candidacy for sequential treatments itself is a prognostic marker and does not confirm survival improvement from any particular sequencing strategy.

First treatment option and CUSE framework per stage

Very early stage (BCLC 0)

This is defined as a single intrahepatic HCC ≤ 2 cm without neoplastic vascular invasion (macrovascular invasion) or extrahepatic spread in a patient with preserved liver function and no cancer-related symptoms.^{42,43} BCLC-0 management varies according to the potential access to LT and specific profiles as described in CUSE BCLC0/A section.

In patients without clinically significant portal hypertension (CSPH; hepatic venous pressure gradient [HVPG] ≥ 10 mmHg or surrogate markers⁴⁴), liver resection (LR) remains the recommended first step.^{45–50} Because of the risk of recurrence, the potential for *Ab initio* LT should always be considered – according to local policy^{51,52} – in selected cases, or reserved as rescue therapy if recurrence occurs. However, regulations and prioritisation rules differ worldwide and may prevent *Ab initio* LT and delay timely LT. Indeed, for very early HCC an alternative approach is to “ablate and wait” until recurrence takes place and at that time consider LT if still within criteria.^{37,53,54}

AFP levels >1,000 ng/ml signal an unacceptably high risk of post-transplant recurrence.^{21,22} If LT is not an option, ablation with percutaneous thermal techniques (radiofrequency [RFA] or microwave [MW]) has historically been the standard. Increasing evidence indicates that TACE, transarterial radio-embolisation (TARE) and external-beam radiotherapy (EBRT) may achieve comparable outcomes. Thus, these modalities

are considered valid alternatives in current practice, with the final choice largely driven by tumour features, expertise, and local resources.

In selected patients, minimally invasive resection (laparoscopic/robotic) may also be considered, particularly for peripherally located tumours where ablation carries higher risks, such as tract seeding or damage to adjacent organs.⁸

Percutaneous ablation offers survival comparable to LR with less invasiveness.^{55–60} The prospective SURF-RCT did not demonstrate a survival advantage for LR over RFA in early, small, predominantly single HCC, supporting ablation as a valid first-line option.⁵⁵ Minimally invasive LR (laparoscopic/robotic) reduces perioperative morbidity vs. open surgery, but its survival impact remains uncertain.⁶¹ Observational signals in very small cohorts conflict with meta-analytic findings, so no definitive superiority can be claimed.^{60,62} MW provides more reliable local control than RFA for tumours up to 3–4 cm and is therefore the preferred percutaneous technique when feasible.^{63,64}

For radiation-based approaches, TARE achieves high objective responses, including frequent complete responses in small lesions.^{63,65} In the recurrent setting post resection/RFA, EBRT shows survival and toxicity comparable to repeat RFA.⁶⁶

Early stage (BCLC-A)

This is defined as a single intrahepatic HCC >2 cm, or up to three HCCs none larger than 3 cm, without macrovascular invasion, extrahepatic spread, and ECOG PS 0 in a patient with preserved liver function. For patients with hepatic decompensation, management principles are discussed in the CUSE 0/A section.

In BCLC-A, treatment selection is guided by HCC-related factors such as tumour burden and location as well as presence of CSPH, and should also consider available expertise and technology.

Single intrahepatic HCC

As in BCLC-0, treatment selection requires a multiparametric approach. Liver function should be stratified by CSPH which predicts higher postoperative risk and worse survival.^{67–69} Non-invasive methods may help to rule out or confirm CSPH but HVPG is still the gold standard.^{44,70} However, HVPG reliability may be suboptimal in steatotic liver disease.^{71,72} HVPG may decrease after eradication of HCV infection, but this has to be carefully assessed.⁷³

In the absence of CSPH, resection (preferably minimally invasive) is the first option. In patients with mildly increased CSPH, minimally invasive surgery remains an option for single peripheral HCC with adequate remnant liver volume.

Surgery is preferred in potential LT candidates, as pathology provides key prognostic data. The presence of microvascular invasion and satellites, predictors of recurrence,^{45–50} may support *Ab initio* LT.^{51,52}

The macrotrabecular-massive pattern is a morphologically distinct and aggressive subtype of HCC but only microvascular invasion and satellite nodules are validated predictors of aggressive recurrence and mortality.⁴⁷ The VETC (vessels-encapsulating tumour clusters) pattern, typically highlighted by CD34 immunostaining, is another predictor of poor prognosis⁷⁴ that often overlaps with the macrotrabecular-massive pattern. Other parameters have been proposed, such as liquid biopsy,^{75,76} to stratify recurrence risk, but only

conventional pathological predictors are validated, and no molecular profile is yet reliable for recurrence risk assessment.

In patients with CSPH, resection poses significant risk,^{68,77} and LT offers better survival. However, when LT is not an option and ablation is suboptimal (e.g. large tumours), alternative treatments may be considered – including intra-arterial locoregional therapies or external beam radiation, or laparoscopic resection in selected cases with favourable tumour location and mild CSPH – before moving to systemic therapy in BCLC-A, consistent with the CUSE framework.^{61,78}

Multifocal up to three HCC nodules, each ≤3 cm

Within Milan criteria (MC) defined by up to three HCC nodules, each ≤3 cm,⁷⁹ multifocal disease is best managed by LT, given the high risk of recurrence after resection or ablation.^{79–81} When LT is not feasible, ablation is the recommended option because the relative merits of resection or intra-arterial locoregional treatment or EBRT vs. RFA/MW remain unsettled.

CUSE tailored approach – BCLC-0/A patients

Treatment decisions for people with very-early or early HCC (BCLC-0/A) often shift when person-specific factors rule out the default option shown in Fig. 5. The literature supporting this proposal is detailed in SUP-A.

If LT (BCLC-0) or surgical resection for single lesions (and LT for multifocal BCLC-A) are not feasible, but liver function is preserved, ablation remains the pillar of care.

For patients listed for LT, oncologic treatment is recommended to prevent HCC progression that could rule out LT, particularly when there is an increased risk of HCC progression or prolonged waiting-list time. Ablation and intra-arterial locoregional treatment are the most widely used options for this purpose.

Patients with single HCC and severe liver dysfunction or decompensation should proceed directly to LT if they meet listing criteria (age, comorbidities, and other relevant factors). When LT is contraindicated for non-HCC reasons, these individuals are classified as BCLC stage D because survival is limited chiefly by liver failure. However, some contraindications may be transient and thus, the multidisciplinary team must assess whether the exclusion factor can be corrected, allowing LT candidacy to be reconsidered if the condition improves and the HCC has not progressed beyond local transplant criteria.

When ablation is selected for a BCLC 0/A patient, choosing the specific modality exemplifies CUSE decision-making: the multidisciplinary team must tailor the technique to the individual's circumstances and to the available evidence for single intrahepatic lesions. The 2026 BCLC update supports a CUSE-guided choice of ablation procedure (RFA/MW)^{82,83} as the preferred percutaneous approach. Laparoscopic ablation is reserved for selected cases, and percutaneous ethanol injection is considered only in rare situations with technical or safety constraints. TARE offers comparable efficacy, while EBRT has demonstrated antitumor activity with median survival exceeding 5 years.^{63,65,84–87}

TARE was endorsed in BCLC 2022 as an ablative alternative, supported by the LEGACY trial (median tumour size 2.6 cm; range 0.9–8.1 cm), which reported a 36-month survival rate of 86.6%,⁶⁵ and more recently validated by the RASER

trial (mean tumour size 2.1 cm, standard deviation 0.4),⁶³ with an actuarial 1- and 2-year OS of 96%. Treatment planning has since been refined, based on dosimetric recommendations for radiation segmentectomy.⁸⁸

Super-selective delivery (“radiation segmentectomy or lobectomy”) has achieved durable responses in retrospective studies, comparable to ablation in patients unsuitable for surgery or percutaneous techniques.^{63,65,89,90} Radiation lobectomy can also increase future remnant liver volume and facilitate resection in borderline cases.⁹¹

EBRT has advanced with improved technology and better understanding of liver and tumour response. Techniques such as stereotactic body radiotherapy (SBRT), proton beam, and carbon ion therapy can deliver ablative doses with a broad therapeutic window. Consensus guidelines now define parameters for ablative EBRT,⁹² with <5% risk of severe toxicity.⁹³ A phase II trial of SBRT for single HCC ≤4 cm unsuitable for resection or RFA reported a 3-year OS rate of 65.4% (95% CI 54.5–75).⁸⁵ Pooled analyses including 484 BCLC-0 and 1,123 BCLC-A patients (134 and 468 treatment-naïve, respectively) showed favourable long-term outcomes: in BCLC-0 patients, median OS was 8.6 years and was not reached in treatment-naïve patients. Median OS in BCLC-A patients was 4.6 years, increasing to 5.4 years in treatment-naïve patients. Subgroup analysis revealed that BCLC-A patients treated with photons had a median OS of 4.3 years, compared with 4.9 years in those treated with protons.⁸⁷ The TRENDY trial,⁸⁴ a multicentre phase II study of patients with single naïve HCC ≤6 cm (median 3.4 cm), reported a median OS of 36.8 months with drug-eluting beads (DEB)-TACE and 44.1 months with SBRT (HR 0.58; 95% CI 0.18–1.85; *p* = 0.36). However, most RCTs enrolled patients with recurrent HCC after prior therapy, and several failed to meet accrual targets,^{94,95} limiting the strength of the evidence for first-line use.

These results suggest encouraging outcomes in early disease, particularly BCLC-0/A with well-preserved liver function, although heterogeneity and selection bias remain important limitations.⁸⁷ For these reasons, BCLC 2026 incorporates EBRT as an ablative option in BCLC-0 patients.

Complete pathological response (CPR) is frequently observed in small solitary tumours after selective or super-selective TACE,^{96–100} but radiologic complete response does not consistently predict histologic necrosis.¹⁰¹

A retrospective analysis of UNOS (United Network for Organ Sharing) liver transplant candidates with an HCC exception (2019–2024) reported that first-line TARE was associated with a higher rate of complete pathologic necrosis on explant compared to TACE (~34% vs. ~20%, respectively) and a lower risk of waitlist dropout (IPTW-adjusted HR 0.78; 95% CI 0.68–0.89).¹⁰²

While higher CPR rates have been reported with certain locoregional therapies – including TARE – SBRT achieved an overall CPR rate of 25% in a series of 88 patients (30% when used for bridging vs. 12% for downstaging).¹⁰³ Thus, CPR may serve as a useful tool for assessing treatment efficacy, but it is not yet a definitive surrogate for OS. Although TACE is not strictly ablative, outcomes from TRENDY and retrospective cohorts suggest OS comparable to SBRT and TARE, and cohort data indicate favourable long-term results in early stages, with apparent advantages in BCLC-0 over BCLC-A, as expected.^{104,105}

While some locoregional therapies are associated with higher rates of CPR, long-term survival differences across

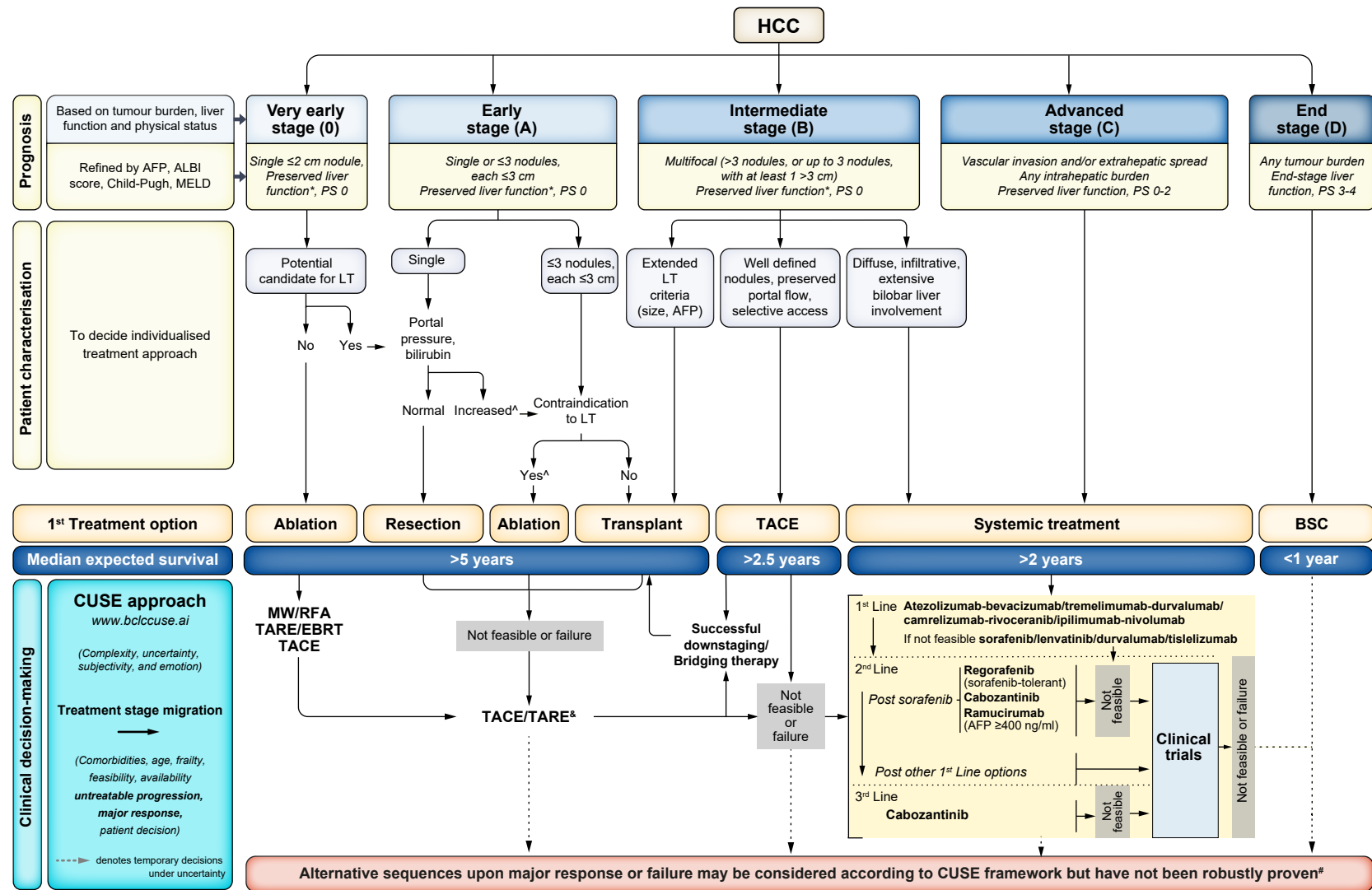


Fig. 5. BCLC staging and treatment strategy in 2026. The BCLC system establishes a prognosis in accordance with the five stages that are linked to first-line treatment recommendation. The expected outcome is expressed as median survival of each tumour stage according to the available scientific evidence. Individualized clinical decision-making, according to the available data on August 5th, 2025, is defined by teams responsible for integrating all available data with the individual patient's medical profile. Note that liver function should be evaluated beyond the conventional Child-Pugh score. AFP, alpha-fetoprotein; ALBI, albumin-bilirubin; BCLC, Barcelona Clinic Liver Cancer; BSC, best supportive care; CUSE approach, Complexity, Uncertainty, Subjectivity, and Emotion factors involved in the clinical decision-making process; EBRT, external beam radiotherapy; ECOG PS, Eastern Cooperative Oncology Group performance status; LT, liver transplantation; MELD, model for end-stage liver disease; ST, systemic therapy; TACE, transarterial chemo-embolisation; TARE, transarterial radioembolisation. *Except for those with tumour burden acceptable for transplant; -Resection for single peripheral HCC with adequate remnant volume should, when feasible, be performed minimally invasively; ^ATACE is classified as non-ablative; ^AEBRT or resection may be considered prior to systemic treatment in BCLC-A; [#]TACE, TARE, EBRT, resection, or LT post-ICI after major response fall in this category. Dotted arrow into the clinical decision-making chapter illustrates that treatment sequences may occasionally bypass systemic therapy; however, this strategy is not a general recommendation and highlights the need for temporary decisions in the face of uncertainty and limited evidence. An interactive version of this figure is available at: https://jhepatology.github.io/Interactive_FlowChart/

modalities remain modest, highlighting the importance of individualised patient selection and multidisciplinary evaluation.^{106–108}

Retrospective cohort comparisons between bland embolisation (transarterial embolisation [TAE]) and DEB-TACE are confounded by selection and centre-strategy biases. These results should be interpreted descriptively, not as definitive head-to-head evidence.

BCLC-A multifocal (up to three lesions, each ≤3 cm)

BCLC 2026 upholds the 2022 recommendations regarding the role of ablation but also integrates additional evidence for tumour board consideration: (a) Vitale *et al.* and Yu *et al.* report favourable outcomes after resection in selected multifocal BCLC-A cohorts;^{109,110} (b) a recent RCT in patients with small, late-recurrence HCC found no survival difference between resection and RFA combined with conventional TACE;¹¹¹ and (c) in BCLC 0/A, emerging data from individual patient data meta-analyses suggest promising long-term outcomes with EBRT.⁸⁷

In HBV-related HCC with CSPH within MC, an open-label RCT¹¹² found no clear survival advantage of partial hepatectomy over locoregional therapy – even within the portal-hypertension subgroup – underscoring that CSPH is the main driver of surgical risk rather than modality-specific benefit. Among trials with uniform early-stage populations, Fang *et al.* (BCLC-A) showed that adding TACE to resection improved outcomes vs. resection alone in selected candidates,¹¹³ whereas Méndez-Romero *et al.* (BCLC-A) reported similar survival between DEB-TACE and SBRT despite early termination of the trial.⁸⁴ For patients with BCLC-A multifocal disease, no single treatment pathway can be established, as therapeutic choice must be tailored to disease characteristics and patient profile.

While resection may appear favourable for early multifocal HCC in terms of absolute survival, comparisons with locoregional therapies remain limited by strong selection bias – including age, number of lesions, liver function, and portal hypertension – despite propensity-score analyses. Moreover, Wang *et al.*¹¹⁴ also reported that MW ablation was associated with fewer complications and shorter hospital stay than resection, underscoring that minimally invasive resection should not be the default first-line option in multifocal BCLC-A but rather weighed through a CUSE evaluation that balances feasibility, safety, oncologic risk, and patient priorities.

Minimally invasive resection/ablation for disease inside the MC is practiced in well-compensated patients, as reflected in the EASL guidelines.¹⁰ However, the supporting evidence is limited – RCTs included very few multifocal cases (e.g. 15 per arm in SURF,¹¹⁵ 24 vs. 31 in IMbrave 050,¹¹⁶ and <10% in Yuan *et al.*¹¹²), while retrospective series involved highly selected patients with CP-A cirrhosis and low tumour burden (mostly two nodules, platelet count >134 × 10³/μl, and <7% requiring major resection). Phase II data for EBRT and TARE remain limited and heterogeneous. Hence, the BCLC 2026 figure includes a CUSE note indicating that resection, TARE and EBRT may be considered in selected cases, ensuring alignment with the EASL guidelines while avoiding unjustified generalisation.

Irreversible electroporation¹¹⁷ and high-intensity focused ultrasound^{118–120} appear to be effective and safe ablation modalities, but robust evidence in HCC is still awaited before formal recommendations can be made. If percutaneous ablation is not feasible, TACE remains the standard first-line option, supported

by the largest body of evidence, whereas resection, EBRT, and TARE are considered only in selected, well-compensated patients under a CUSE framework.

Role of laparoscopic approaches, age and HCC location

Recent studies have compared laparoscopic approaches and ablation techniques in early-stage HCC. In a multicentre analysis, laparoscopic MW ablation showed lower morbidity than laparoscopic hepatectomy (14.7% vs. 34.7%; *p* = 0.006), while OS at 1, 3, and 5 years was similar between both groups (laparoscopic MWA: 88.2%, 69.9%, 45.6% vs. hepatectomy: 86.1%, 72.9%, 51.4%; *p* = 0.693).^{121,122} In patients with portal hypertension, the complication rate of MW was significantly lower than that of liver resection (*p* <0.001), supporting ablation as a safer option in this setting.¹²³

Tumour location is also a key determinant factor in treatment selection. Indeed, HCC location and patient age are key factors often considered by multidisciplinary boards when deciding treatment. However, data on their impact after ablation remain limited. In a recent cohort,¹²⁴ neither tumour location nor age (elderly vs. non-elderly) at the time of first ablation were associated with OS or with the need to change treatment strategy.^{125–130}

By contrast, in a cohort of 225 patients treated with RFA/MW at the BCLC group, suboptimal locations (subcapsular, dome, proximity to heart, hilum, gastrointestinal tract, or major vessels) were linked to shorter time to recurrence (13.1 vs. 20.7 months) and earlier need for treatment change (17.6 vs. 25.1 months) compared with optimal sites.¹²⁴ Taken together, these findings suggest that while tumour location may not independently impact survival, ablation in technically challenging sites carries a higher risk of earlier recurrence and strategy modification.

Functional decline and frailty significantly influence peri-operative outcomes in elderly patients with HCC, regardless of surgical approach. While minimally invasive hepatectomy reduces hospital stay and morbidity, these factors remain key predictors of complications. The Barthel index has been shown to independently predict postoperative risk, underscoring its value in preoperative assessment.¹³¹

Intermediate stage (BCLC-B)

In the intermediate stage, tumour burden is defined as more than three HCC nodules, or up to three nodules with at least one larger than 3 cm and ECOG PS 0. Preserved liver function is required to be considered for locoregional therapies or down-staging strategies. Patients with hepatic decompensation enter the pathway of decompensated cirrhosis (Fig. 2). They may be evaluated for LT if they meet local listing criteria. The 2026 update leaves the BCLC-B stratification unchanged from 2022⁸ as no new high-quality evidence warrants modification.

The first subgroup includes patients with well-defined HCC nodules who may be candidates for LT if they meet the institution's *Extended LT* criteria. AFP level >1,000 ng/ml signals an unacceptably high risk of post-transplant recurrence.^{21,22} Where access to LT is limited by donor shortage, living-donor grafts may mitigate this constraint.^{132,133} Such an alternative must balance donor risk against community benefit, underscoring the need for expert, CUSE-guided judgment.

The second subgroup comprises patient's ineligible for LT but with preserved portal flow and defined tumour burden,

allowing selective access to feeding tumour arteries; these individuals are candidates for intra-arterial therapies. TACE remains the standard locoregional therapy, as no head-to-head trials have proven any alternative superior in terms of OS. If tumour distribution or hepatic function precludes effective TACE, the pathway proceeds directly to systemic therapy, with detailed consideration of potential alternatives as outlined in the CUSE section. AFP levels, subtle gradients of liver dysfunction within CP-A, and tumour heterogeneity all modulate prognosis, yet robust universal thresholds remain elusive. Several scores have been proposed to predict prognosis after TACE (SUP-B). However, they do not alter TACE candidacy under conventional selection criteria.

The third subgroup includes patients with diffuse, infiltrative, or extensive HCC liver involvement, in whom selective TACE is not feasible. In such cases, systemic therapy is recommended, although no strict cut-off defines this transition.

Treatment for BCLC-B

Conventional TACE (cTACE) involves injection of a chemotherapy-lipiodol emulsion followed by vessel occlusion with gelatin sponge.¹³⁴ DEB-TACE enables slow drug release, reducing systemic peaks while maintaining efficacy and improving tolerability.^{135,136} A higher incidence of biliary changes has been reported with DEB-TACE,^{137,138} though these are mostly image-detected dilatations without clinical impact.

Two randomised trials have evaluated the impact of DEB-TACE compared with TAE in HCC. Malagari *et al.* demonstrated DEB-TACE achieved higher complete response rates (26.8% vs. 14%) and reduced early recurrence at 12 months (45.7% vs. 78.3%) in patients with intermediate-stage HCC, suggesting a benefit in local tumour control.¹³⁹ In contrast, Brown *et al.* found no significant differences in objective response rate, PFS, or OS between DEB-TACE and bland embolisation in an HCC population beyond BCLC-B.¹⁴⁰ This broader selection may help explain the suboptimal survival. Thus, the trial does not establish equivalence between the two options. In a cohort of 1,461 patients with HCC (730 treated with TACE and 731 with TAE), the TAE group had poorer liver function and higher HBV prevalence.¹⁴¹ Propensity-matched analyses favoured TACE over TAE for OS and response. In contrast, Roth *et al.*¹⁴² found that an apparent survival advantage of TACE attenuated after propensity weighting in the overall cohort. However, among CP-A, TACE retained a survival advantage, suggesting that benefit is concentrated in compensated patients.

The STOPPER trial¹⁴³ showed high short-term survival with epirubicin-loaded TACE, and TACTICS showed that outcomes with TACE alone were more numerically favourable within the up-to-7 criteria score¹⁴⁴ than outside. Collectively, these findings indicate that the efficacy of TACE alone depends heavily on tumour burden and BCLC stage distribution. In summary, when liver function is preserved, TACE is generally preferred over TAE (see SUP-A). In unselected or less compensated populations, the difference is less certain, and residual confounding limits causal inference.

In practice, patients with extensive bilobar disease are unlikely to benefit from TACE, making systemic therapy the preferred option. The transition between these subgroups is best guided by the CUSE framework, balancing feasibility, risk, and patient priorities. However, available evidence lacks

granular data on systemic therapy outcomes in this subgroup, as baseline trial characteristics have not reported such detail; further clarification is awaited. The CUSE section offers further insight into treatment selection for BCLC-B patients.

CUSE tailored approach – BCLC-B patients

Treatment decisions for people with intermediate HCC often shift when person-specific factors rule out the default option shown in Fig. 5. The literature supporting this proposal is also detailed in SUP-A.

While the MC⁷⁹ are the standard benchmark, growing evidence supports the inclusion of certain BCLC-B patients beyond MC – particularly those successfully downstaged and with AFP levels below a specific cut-off that varies across jurisdictions – who can achieve post-LT outcomes equivalent to those within MC.^{145–147} This has led to the development of extended and biomarker-integrated selection models. In contrast, AFP levels exceeding 1,000 ng/ml consistently predict poor survival and higher recurrence risk, and are widely considered a contraindication to LT, regardless of tumour burden.^{21,22}

Although AFP reduction following downstaging is common, the optimal magnitude and duration of decline required for transplantation eligibility remain undefined in this setting, though it must be weighed carefully against donor risk and overall societal benefit.^{133,148}

In this context, what is often termed “neoadjuvant” therapy largely corresponds to downstaging or bridging therapy, widely applied in patients beyond MC but within extended criteria to prevent drop-out and refine selection. LT after major response to systemic treatment – particularly immunotherapy – is also emerging, though still investigational and supported only by limited prospective evidence. To avoid ambiguity, we refer to these strategies as “bridging therapy” and “systemic treatment prior to LT.” Both remain within the realm of clinical decision-making and, in the 2026 BCLC, systemic treatment prior to LT is categorized as *alternative sequences that may be considered but have not been proven*, underscoring the persistent uncertainty surrounding LT indications.

Resource allocation requires balancing individual benefit with societal utility.¹⁴⁹ LT should be prioritised for candidates achieving the largest net survival gain over alternatives, without excluding those who may derive substantial, though not maximal, benefit. Modern utility frameworks incorporate dynamic parameters, including tumour control duration, AFP kinetics, and overall trajectory on the waitlist, to align efficiency with equity in organ distribution.¹³³

When LT is not an option, patients with preserved function should receive TACE as a default, supported by RCT evidence.^{136,150} Conventional and DEB techniques provide comparable tumour control,¹⁵⁰ with survival estimates coming largely from cohort and phase II studies.^{84,151–153} The STOPPER trial enrolled candidates with BCLC-A/B disease and preserved liver function to receive TACE.¹⁴³ The TACTICS trial enrolled patients across BCLC stages A, B, and C, with the majority being BCLC-B, with smaller subsets representing early and selected advanced stages.¹⁵⁴

If TACE is not feasible due to excessive tumour burden, systemic therapy should be considered. Alternatively, if TACE is not performed for technical reasons rather than tumour burden,

other locoregional options such as TARE or EBRT may be used to delay systemic therapy and preserve it for progression.

Although retrospective studies showed higher tumour response and longer time to progression with TARE vs. TACE in this population, no survival benefit difference has been demonstrated.^{155–157} TRACE, the only randomised trial, included patients with ascites and ECOG-PS 1, which contributed to early mortality in the TACE arm and raised concerns about the study's validity.⁹⁰ OS in the TACE arm was unexpectedly low – close to the natural history of BCLC-B¹⁵⁸ – even though the study also included BCLC-A cases, which limits the robustness and clinical interpretation of these findings.

More reliable benchmarks come from larger non-randomised cohorts: in 1,000 patients treated with TARE,¹⁵⁹ median OS for BCLC-B was 19 months overall, but 38.5 months in patients with ALBI grade 1, underscoring the prognostic relevance of liver function. These results suggest a benchmark survival of 3 years¹⁶⁰ for patients with well-compensated BCLC-B treated with TARE.

Separately, a meta-analysis of three prospective randomised trials by Kuehnle *et al.*¹⁶¹ comparing EBRT with TACE found no difference in OS or grade ≥ 3 toxicity, but applicability to BCLC-B is constrained by small stage-specific numbers and heterogeneity in treatment modalities (protons vs. photons for EBRT; cTACE, DEB-TACE, and TAE on the embolisation side). Encouraging results have also emerged for EBRT in BCLC-B. Moon *et al.*⁸⁷ reported favourable OS in an unselected cohort, with particularly long survival in treatment-naïve patients and those with ALBI grade 1 (4.5 years). However, unclear selection criteria limit generalisability and hinder identification of patients most likely to benefit. Sorin *et al.* likewise reported favourable multi-year survival with EBRT in BCLC-B,¹⁶² but these retrospective, small-cohort data are best considered descriptive rather than comparative.

In summary, OS data on TAE are not robust enough to replace TACE in well-selected BCLC-B patients. TARE or EBRT may be considered for patients who are not candidates for TACE, but no data support its use as a first-line option based on OS. If considering TARE, the DOSISPHERE-01 trial showed that personalised dosimetry improves outcomes and should be standard practice.¹⁶³

The current BCLC strategy still does not support TARE or SBRT as first-line alternatives to TACE in this context, as the available data on these therapies in non-TACE candidates primarily involve patients with BCLC-A disease. These modalities could serve to delay the initiation of systemic therapy and preserve it as a subsequent option in the event of disease progression.

Treatment stage migration

While outcomes are commonly stratified by liver function or prior therapy, few studies address individual profiles (e.g. BCLC-B/C with diabetes or other comorbidities). This gap underscores the need for granular data to guide therapy beyond OS-based choices. CUSE highlights the importance of considering the reason for TSM (Fig. 3), ensuring that subsequent decisions reflect evolving clinical status and patient goals. Fig. 5 (dotted arrow) illustrates that sequences may occasionally bypass systemic therapy; however, this is not a general recommendation and reflects temporary decisions in settings of uncertainty.

TACE-based combinations

Phase III trials have evaluated TACE plus systemic therapy in unresectable HCC. Multiregional studies evaluating immunotherapy combinations (EMERALD-1¹⁶⁴, LEAP-012¹⁶⁵, and TALENTACE¹⁶⁶) demonstrated improvements in PFS (HR 0.64–0.77); however, OS data remain immature, with stage-specific outcomes reported only in TALENTACE. Recently, a sponsor press release reported interim findings from LEAP-012, indicating that the combination is unlikely to demonstrate a statistically significant OS benefit compared with TACE alone. <https://us.eisai.com/press-releases/merck-and-eisai-provide-update-on-phase-3-leap-012-trial-in-unresectable-non-metastatic-hepatocellular-carcinoma> Three large RCTs in China – LAUNCH, (lenvatinib + TACE)¹⁶⁷ TACE + apatinib,¹⁶⁸ and DEB-TACE + apatinib¹⁶⁹ – each showed OS and PFS benefit, but lack validation in other populations. Differences in trial design, BCLC composition, and geographic coverage preclude firm conclusions. At present, TACE-based combinations cannot be recommended outside clinical trials. This exemplifies the value of CUSE in contextualizing signals of efficacy against heterogeneity, scope, and evidence maturity (SUP-C). Beyond these specific trials, we stress that interpretation of immature or heterogeneous evidence requires a structured approach. As detailed in SUP-A (BCLC CUSE 2026 literature interpretation), we advocate contextualising these signals of efficacy within MTBs, weighing safety, feasibility, and patient values before recommending implementation outside clinical trials.

Advanced stage (BCLC-C)

This stage includes patients with macrovascular invasion and/or extrahepatic spread, regardless of intrahepatic tumour burden, which in Fig. 5 is indicated as “any intrahepatic burden” to denote its coexistence with macrovascular invasion or extrahepatic spread. They have preserved liver function and a maintained performance status (ECOG PS 0–2) and should be considered for systemic therapy.

For first-line treatment, immune-based combinations – atezolizumab plus bevacizumab,¹⁷⁰ tremelimumab plus durvalumab,¹⁷¹ ipilimumab plus nivolumab,¹⁷² and camrelizumab plus rivoceranib (with only 17% of patients from non-Asian countries)¹⁷³ – have all demonstrated superior OS compared to sorafenib (or sorafenib/lenvatinib in the case of ipilimumab plus nivolumab) across multiple geographic regions (SUP-B).

These trials included patients with unresectable HCC, mostly stage BCLC C but also A and B in variable proportions, which contributed to differences in outcomes. Other treatments such as lenvatinib,¹⁷⁴ durvalumab¹⁷¹ or tislelizumab¹⁷⁵ were non-inferior to sorafenib,^{176,177} which itself improved OS vs. placebo.

Other regimens (SUP-C) such as anlotinib plus penpulimab,¹⁷⁸ HAIC-FOLFOX,^{179,180} donafenib,¹⁸¹ TACE plus lenvatinib,¹⁶⁷ sintilimab plus bevacizumab biosimilar,¹⁸² and toripalimab plus bevacizumab¹⁸³ demonstrated OS benefit over sorafenib or lenvatinib. However, these RCTs were conducted exclusively in Asia (primarily China).

Phase III trials (SARAH,¹⁸⁴ SIRVeNIB,¹⁸⁵ and SORAMIC¹⁸⁶) comparing TARE alone or in combination with sorafenib vs. sorafenib monotherapy were negative, and by the time immunotherapy became the standard first-line treatment, no randomised trial had demonstrated a survival benefit for TARE

in this setting. Additionally, several studies have assessed whether the degree of vascular invasion should inform treatment selection. A recent systematic review and meta-analysis of 73 studies (10,329 patients with macrovascular invasion) further evaluated this subgroup. Despite a rigorous identification process, substantial heterogeneity persisted for most modalities ($I^2 > 80\%$), precluding robust pooling. TARE was the only treatment for which 1-year OS was the most comparable endpoint across modalities, achieving a pooled 1-year OS of 34%, with wide confidence intervals (95% CI 2–48%; $I^2 = 0\%$). In contrast, surgery, TACE, EBRT, and sequential strategies were confounded by indication and immortal-time bias, and survival data beyond 12 months were largely incomplete.¹⁸⁷

These findings underscore that systemic therapy remains the standard for BCLC-C, while TARE may represent an alternative in selected patients when systemic therapy is contraindicated or not feasible, highlighting the need for stratified head-to-head trials of it against immunotherapy. Because no head-to-head trials directly compare the available immunotherapy-based combinations, current guidance is to select one as initial systemic therapy, tailoring this choice through the CUSE framework.

Systemic treatment sequence

Among first-line therapies, only sorafenib^{176,177} supports an evidence-based sequencing strategy, as validated pathways following other first-line regimens are lacking.

Patients progressing to the second-line may benefit from regorafenib,^{188,189} cabozantinib regardless of prior sorafenib tolerance,¹⁹⁰ or ramucirumab if AFP >400 ng/ml,¹⁹¹ irrespective of prior tolerance. Cabozantinib has also demonstrated efficacy as a third-line option.¹⁹⁰ In the field of immunotherapy, data are limited. A Western trial of pembrolizumab vs. placebo in the second line did not meet its primary OS endpoint,¹⁹² whereas an Asian trial in a similar population reported a significant OS improvement.¹⁹³ The recently published meta-analysis including both trials found that pembrolizumab improved OS (HR 0.79, 95% CI 0.67–0.93).¹⁹⁴ However, the control arms differed slightly, as the Asian trial also included patients who had progressed on or were intolerant to oxaliplatin-based chemotherapy.¹⁹³

As mentioned, the optimal sequence following immune checkpoint inhibitor therapy in BCLC-C is not established: non-randomised data and indirect comparisons suggest tyrosine kinase inhibitor activity after immunotherapy without proven superiority of any sequence. We recommend prioritising preservation of liver function, tolerability, and clinical-trial enrolment. SUP-B&C describe outcomes according to different treatment options.

CUSE-tailored approach in BCLC-C patients

As of August 5, 2025, SUP-B summarises available data to guide first-line decisions. In brief, five positive multiregional RCTs and more than 30 sub-analyses of pivotal studies have characterised populations, updated OS, or described quality of life. Over 1,400 real-world publications are also available, although cohort overlap cannot be reliably excluded. Four RCTs conducted in China have reported encouraging results. However, OS benefits have not been

demonstrated in multiregional populations, precluding a general recommendation.

Regarding second-line therapy, RCTs are available only for those patients who received sorafenib as first-line treatment.^{188,190,191} No phase III multiregional RCT has evaluated patients previously treated with immune-based regimens. In total, six phase II trials have analysed outcomes with cabozantinib,^{195,196} regorafenib,¹⁹⁷ regorafenib-nivolumab,¹⁹⁸ regorafenib-pembrolizumab,¹⁹⁹ and lenvatinib.²⁰⁰ Finally, >30 real-world studies describe outcomes in this population. See SUP-B for details on real-world studies by country, specific condition, and treatment line.

First-line therapy selection according to CUSE 2026

Selecting first-line therapy requires balancing multiple elements: evidence from pivotal trials (eligibility criteria and survival benefits);^{170–173} safety data from trials and real-world studies; patient comorbidities; access to treatment;²⁰¹ expected response and progression rates including progression patterns;^{28,189,202–207} long-term outcomes; biological rationale; and quality-of-life metrics, including the granularity of outcome measures to predict the individual course (SUP-B).

When more than one regimen has shown survival benefit, pragmatic considerations – such as time toxicity,²⁰⁸ indirect costs,²⁰⁹ patient experience²¹⁰ – may influence the choice, although such analyses are rarely reported in clinical trials or post-approval studies, including patient preference data.²¹¹ In the absence of head-to-head comparisons or when availability is limited, decisions should be individualised. Within the CUSE framework, treatment selection is guided by predefined priorities that extend beyond OS to encompass safety, objective response rate, PFS, durable response, and other outcomes. Because no universal hierarchy exists, the multidisciplinary team and the patient should first agree on primary goals. Accordingly, the BCLC model emphasises assembling the full safety–efficacy dataset and weighting endpoints within a CUSE-guided discussion, ensuring that the final plan reflects both clinical judgment and the goals and values of the person living with HCC.

A profile of special interest within BCLC-C is vascular invasion. Isolated cases are extremely rare;^{187,212,213} the typical presentation is intrahepatic HCC with concomitant macrovascular invasion. Although immunotherapy is the general recommendation for BCLC-C, current evidence in patients with vascular invasion comes mainly from subgroup analyses of pivotal trials²¹⁴ and observational cohorts,^{212,213} particularly Japanese series with atezolizumab–bevacizumab.^{213,215} These studies confirm the prognostic impact of liver function and show heterogeneous survival depending on invasion extent, but cohort heterogeneity and high rates of missing long-term data limit generalisability. Comparisons with TARE suggest lower 12-month survival, yet wide confidence intervals and divergent populations preclude indirect inference.¹⁸⁷

The DOSISPHERE-01 phase II trial further informs this setting. Patients were randomised (1:1) to personalised dosimetry (≥ 205 Gy to the index lesion, up to 250–300 Gy when feasible) or standard dosimetry (120–150 Gy to the treated volume). In the portal tumour invasion subgroup ($n = 39$), median OS was 22.0 months (95% CI 10.3–36.5) with personalised dosimetry ($n = 18$) vs. 9.4 months (95% CI

5.3–17.6) with standard dosimetry ($n = 21$) (HR 0.52; 95% CI 0.26–1.01; $p = 0.058$). *Post hoc* analyses confirmed a dose–response effect in this subgroup of patients: median OS was 22.9 months (95% CI 11.0–48.1) with tumour doses ≥ 205 Gy vs. 10.3 months (95% CI 5.9–17.6) with doses < 205 Gy (HR 0.42; 95% CI 0.22–0.81; $p = 0.010$).²¹⁶

There is increasing interest in EBRT in patients with vascular invasion in combination with systemic therapy. A seminal phase III RCT (RTOG 1112) compared sorafenib to sorafenib with SBRT, with patients stratified by the presence and degree of vascular invasion.²¹⁷ Of the 177 patients eligible for analysis, 74% had macrovascular invasion and results showed that the vascular HCC response was 9% (95% CI 2–16) with sorafenib alone compared to 38% (95% CI 26–49) with sorafenib plus SBRT ($p < 0.001$). This data refers exclusively to vascular response and is presented for descriptive purposes. Of note, OS and PFS remain the main endpoints of the trial.

These favourable results from prospective trials suggest EBRT can be considered in appropriate patients with macrovascular invasion and should be studied in future trials that incorporate modern systemic therapies.

In addition to the meta-analysis by Fortuny *et al.*¹⁸⁷ on the role of surgery in this population, observational series consistently reflect treatment-selection bias:^{213,215} apparent advantages of resection largely disappear once disease burden, liver function, and propensity-adjusted methods are accounted for. Overall, in the absence of RCT evidence, survival differences across modalities must be interpreted with caution, acknowledging residual confounding and the non-exchangeability of treatment cohorts. Systemic therapy is the preferred first-line option for patients with macrovascular invasion (Vp1–Vp4). If contraindicated or not feasible, TARE (DOSISPHERE-01)^{163,216} or EBRT (RTOG 1112)²¹⁷ may be considered within the CUSE framework; however, evidence in patients with portal vein invasion remains limited, with only 18 such cases included in DOSISPHERE-01, and these options are therefore not yet robustly proven.

Evaluating immunotherapy suitability and management

Assessing suitability for immunotherapy – and identifying absolute contraindications – requires thorough pretreatment evaluation of the risk for irAEs. Equally critical is prompt recognition and management of irAEs^{25,218,219} throughout care, including admission to the intensive care units.²²⁰ This underscores the need for targeted education of both health professionals and persons living with HCC. Specialised oncology nurses play a pivotal role in patient teaching, monitoring, and care coordination.^{25,221,222,267} The principal drivers of systemic therapy management are therapy-related adverse events, complications of cirrhosis, progression of HCC with or without decompensation, and patient adherence (SUP-B). In patients with cirrhosis, careful differential diagnosis of irAEs and other non-immune-related causes should guide cautious and rational use of steroids. Extrapolating recommendations from non-cirrhotic populations may increase the risk of decompensation or bacterial infections.²¹⁸ Additional challenges arise in special populations – people living with HIV,^{223–226} candidates for LT,^{133,227–231} or patients on dialysis.^{232,233} Given the limited evidence base, universal recommendations cannot be made. In patients with cirrhosis,

careful use and appropriate tapering of corticosteroids are required when managing irAEs to minimise the risk of decompensation.

Second- and later-line selection according to CUSE 2026

In the second-line setting and beyond, clinical trial enrolment remains the preferred option for patients with HCC. When trial participation is not feasible, treatment selection must be guided by the CUSE framework. Robust evidence for sequential therapy exists primarily after sorafenib.^{188,190,191} Recommendations after immunotherapy or lenvatinib are based on phase I/II trials, cohort studies, and expert consensus. Alternative strategies – including immunotherapy rechallenge, tyrosine kinase inhibitors, locoregional therapies (TARE, EBRT), or phase II-based combinations – may be considered, but are not validated as standard second-line options (see SUP-C).

Key uncertainties arise when discontinuing first-line therapy without a defined subsequent strategy, such as stopping after complete response, escalating to ablative intent (ablation or surgery), or continuing despite progression or toxicity.²³⁴ Decisions should weigh clinical status, tolerance, and patient goals (see SUP-C). In patients with absolute contraindications or no access to systemic therapy, non-systemic approaches (TARE, EBRT) may be considered within CUSE despite limited evidence.

Downsizing, downstaging and conversion approach through locoregional treatment

Effective treatments can reduce tumour burden, yielding complete or partial responses according to different criteria (RECIST 1.1²³⁵ EASL,²³⁶ RECICL,²³⁷ mRECIST,²³⁸ Choi criteria,²³⁹ iRECIST,²⁴⁰ imRECIST²⁴¹ and LIRADS²⁴²). In such cases, follow-up staging may reveal that tumour downsizing translates into downstaging, allowing reconsideration of treatment options that were initially deemed unsuitable. Downstaging via locoregional approaches (TACE and TARE) has been developed and validated for LT.^{108,147} Patients who initially exceed listing criteria but respond to therapy and fulfil downstaging criteria may achieve optimal outcomes with transplantation. The UNOS downstaging criteria are the most validated (single 5–8 cm, 2–3 lesions ≤ 5 cm with total ≤ 8 cm, or 4–5 lesions ≤ 3 cm with total ≤ 8 cm, plus AFP $< 1,000$ ng/ml).^{108,243} The XXL RCT showed superior 5-year OS rates with LT vs. best alternative therapy after successful downstaging (77.5% vs. 31.2%; HR 0.32; $p = 0.035$).¹⁴⁷ The key issues to address are the upper limits of tumour burden to entry into a downstaging protocol aimed at achieving successful transplantation, and the post-treatment criteria (Milan, UCSF) that must be met for listing.²⁴⁴ Optimal tumour/AFP thresholds, response duration, and definitions of success remain unsettled and preclude universal recommendations.²⁴⁵

Downsizing, downstaging and conversion approach with systemic treatment

Beyond locoregional strategies, systemic therapies are being investigated as tools to achieve tumor regression and potential downstaging. Downstaging with immunotherapy has recently been explored, and early data are promising.^{246,247} Prospective studies are needed to define intention-to-treat outcomes and safety.¹³³ Immune checkpoint inhibitor exposure could

increase rejection risk, particularly with a washout <30 days. A meta-analysis suggests at least a 3-month interval.²⁴⁸ Details of individual clinical reports are summarised in SUP-A&B&C.

The concept of a conversion approach – using systemic or combined treatment to achieve potential cure after major response – remains unproven and is categorised as an alternative sequence in BCLC 2026. Most publications are case reports, with only eight phase II trials and retrospective series.^{249–256} Reported conversion rates vary (2–60%), reflecting heterogeneity in criteria, response assessment, and sequencing. Prospective studies are needed to define baseline thresholds, treatment strategies, response evaluation, and timing of clinical decisions.

End-stage (BCLC-D)

Individuals with HCC who present with major cancer-related symptoms (ECOG PS >2) or severe liver dysfunction and are not candidates for LT due to tumour burden or non-oncologic contraindications have poor short-term survival.^{1,257}

In selected scenarios, particularly when decompensation is potentially reversible – such as HBV-related cirrhosis or alcohol-induced cirrhosis with sustained abstinence – patients may achieve hepatic recompensation,^{258–260} defined per Baveno VII criteria.⁴⁴ Studies have shown that approximately 15–20% of patients with alcohol-related cirrhosis achieve recompensation following sustained abstinence, translating into a marked reduction in liver-related mortality.^{261,262} In HBV-related disease, antiviral therapy can induce recompensation in a substantial proportion of patients.²⁶³

Recent data in patients with HCV report recompensation rates ranging from 36% to 64% across cohorts.^{264,265} Achieving recompensation is associated with improved prognosis and, in some cases, may enable re-evaluation for systemic therapy or LT. Prospective studies should define the extent to which the tumour may progress while awaiting recompensation.

In practice, only 7% of patients meet Baveno VII criteria of recompensation, supporting our statement that most BCLC-D patients will not achieve recompensation. Even when considering expanded criteria, only up to 37% of patients qualified, none of whom had HCC.¹⁹ In this group, tumour-directed therapy provides no meaningful survival benefit. Therefore, the priority shifts to symptom control – for example, radiotherapy for metastases²⁶⁶ – along with supportive measures and coordinated palliative care.

CUSE-tailored approach in BCLC-D patients

Patients receiving best supportive care represent a heterogeneous group, as some patients in this group have earlier-stage disease that is managed conservatively due to comorbidities or patient choice. Clear distinction is essential: whereas patients fulfilling the strict definition of BCLC-D have a prognosis of median OS <3 months, others may experience longer survival (SUP-B). Hence, the expected median OS for BCLC-D in Fig. 5 has been revised to <12 months.

Conclusion

The 2026 BCLC update reflects the challenge of managing HCC within an expanding yet uneven evidence base. CUSE addresses this need by providing open-access staging and treatment support that adapts to evolving data. This version preserves the simplicity of BCLC staging while embedding CUSE to guide individualised care in today's complex therapeutic landscape. As a conceptual framework, CUSE implementation may vary across settings with different resources. Its strength is in making uncertainty explicit and creating a shared language, while future studies should assess its impact on clinical practice and outcomes. To support implementation, the open-access platform (www.bclccuse.ai) has been developed as a decision-support tool, grounded in the CUSE framework and designed to standardize the decision-making process.

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Abbreviations

AFP, alpha-fetoprotein; ALBI, albumin–bilirubin; BCLC, Barcelona Clínic Liver Cancer; CP, Child–Pugh; CPR, complete pathological response; CUSE, Complexity, Uncertainty, Subjectivity, Emotion; DEB-TACE, drug-eluting beads transarterial chemoembolisation; EBRT, external beam radiotherapy; ECOG PS, Eastern Cooperative Oncology Group performance status; HCC, hepatocellular carcinoma; HVPG, hepatic venous pressure gradient; IPTW, inverse-probability of treatment weighting; irAEs, immune-related adverse events; LR, liver resection; LT, liver transplantation; MC, Milan criteria; MELD, model for end-stage liver disease; MTB, multidisciplinary tumour board; MW, microwave ablation; OS, overall survival; PFS, progression-free survival; RCT, randomised controlled trial; RFA, radiofrequency ablation; SBRT, stereotactic body radiation therapy; TACE,

transarterial chemoembolisation; TAE, transarterial embolisation; TARE, transarterial radioembolisation; TSM, treatment stage migration; UTP, untreatable progression.

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Conflict of interest

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Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

A list of the authors' contributions to the study; concept and design MSZ, AF, JB, MR, experiments and procedures MSZ, AF, JB, MR, JR, JF, MB, AGC, AD, NLL, GI, MM, CA; all the authors contributed to writing the article.

Data availability

The 2026 BCLC treatment strategy (cut-off August 5, 2025) builds on prior iterations to guide individualized decision-making. To generate the evidence synthesis, we conducted a PubMed search restricted to humans, adults (≥19 years), and English-language records. We performed (1) an individual-participant-data (IPD) meta-analysis of external beam radiotherapy (EBRT) that included approximately 5,000 patients (Moon *et al.*), and (2) an aggregate-data meta-analysis comprising 73 studies of patients with macrovascular invasion (MVI) (Fortuny *et al.*). We also reviewed all additional relevant information according to prespecified criteria that is described in [Supplementary material A](#) including the role of nurses according to BCLC approach.²⁶⁷

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English Editing Assistance: During the preparation of this work, the author(s) used *OpenAI's ChatGPT (GPT-5, subscription version, temporary session)* to improve the fluency of the English language and enhance conciseness. The final version of the manuscript was reviewed using this tool. After its use, the author(s) carefully revised and edited the content, and take full responsibility for the final text.

Supplementary data

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References

Author names in bold designate shared co-first authorship

- [1] Llovet J, Brú C, Bruix J. Prognosis of hepatocellular carcinoma: the BCLC staging classification. *Semin Liver Dis* 1999;19:329–338. <https://doi.org/10.1055/s-2007-1007122>.
- [2] Llovet JM, Burroughs A, Bruix J. Hepatocellular carcinoma. *Lancet* 2003;362:1907–1917. [https://doi.org/10.1016/S0140-6736\(03\)14964-1](https://doi.org/10.1016/S0140-6736(03)14964-1).
- [3] Forner A, Reig ME, Rodríguez de Lope C, Bruix J. Current strategy for staging and treatment: the BCLC update and future prospects. *Semin Liver Dis* 2010;30:61–74. <https://doi.org/10.1055/s-0030-1247133>.
- [4] Forner A, Llovet JM, Bruix J. Hepatocellular carcinoma. *Lancet* 2012;379:1245–1255. [https://doi.org/10.1016/S0140-6736\(11\)61347-0](https://doi.org/10.1016/S0140-6736(11)61347-0).
- [5] Forner A, Reig M, Bruix J. Hepatocellular carcinoma. *Lancet* 2018;391:1301–1314. [https://doi.org/10.1016/S0140-6736\(18\)30010-2](https://doi.org/10.1016/S0140-6736(18)30010-2).
- [6] Reig M, Darnell A, Forner A, Rimola J, Ayuso C, Bruix J. Systemic therapy for hepatocellular carcinoma: the issue of treatment stage migration and registration of progression using the BCLC-refined RECIST. *Semin Liver Dis* 2014;34:444–455. <https://doi.org/10.1055/s-0034-1394143>.
- [7] Bruix J, Reig M, Sherman M. Evidence-based diagnosis, staging, and treatment of patients with hepatocellular carcinoma. *Gastroenterology* 2016;150:835–853. <https://doi.org/10.1053/j.gastro.2015.12.041>.
- [8] Reig M, Forner A, Rimola J, Ferrer-Fàbrega J, Burrel M, Garcia-Criado Á, et al. BCLC strategy for prognosis prediction and treatment recommendation: the 2022 update. *J Hepatol* 2022;76:681–693. <https://doi.org/10.1016/j.jhep.2021.11.018>.
- [9] Orlandi F, Christensen E. A consensus conference on prognostic studies in hepatology. *J Hepatol* 1999;30:171–172. [https://doi.org/10.1016/S0168-8278\(99\)80023-3](https://doi.org/10.1016/S0168-8278(99)80023-3).
- [10] European Association for the Study of the Liver. EASL Clinical Practice Guidelines on the management of hepatocellular carcinoma. *J Hepatol* 2025;82:315–374. <https://doi.org/10.1016/j.jhep.2024.08.028>.
- [11] Singal AG, Llovet JM, Yarchoan M, Mehta N, Heimbach JK, Dawson LA, et al. AASLD practice guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma. Publish Ah; 2023. <https://doi.org/10.1097/hep.0000000000000466>.
- [12] Nault J-C, Boubaya M, Wartschi M, Dohan A, Pol S, Pop G, et al. [18F] fluorodeoxyglucose and [18F]fluorocholine PET-CT for staging

- optimisation and treatment modification in hepatocellular carcinoma (PET-HCC01): a prospective multicentre study. *Lancet Gastroenterol Hepatol* 2025;10:306–314. [https://doi.org/10.1016/S2468-1253\(25\)00011-1](https://doi.org/10.1016/S2468-1253(25)00011-1).
- [13] Johnson PJ, Berhane S, Kagebayashi C, Satomura S, Teng M, Reeves HL, et al. Assessment of liver function in patients with hepatocellular carcinoma: a new evidence-based approach—the ALBI grade. *J Clin Oncol* 2015;33:550–558. <https://doi.org/10.1200/JCO.2014.57.9151>.
- [14] Pinato DJ, Sharma R, Allara E, Yen C, Arizumi T, Kubota K, et al. The ALBI grade provides objective hepatic reserve estimation across each BCLC stage of hepatocellular carcinoma. *J Hepatol* 2017;66:338–346. <https://doi.org/10.1016/j.jhep.2016.09.008>.
- [15] Kamath PS, Wiesner RH, Malinchoc M, Kremers W, Therneau TM, Kosberg CL, et al. A model to predict survival in patients with end-stage liver disease. *Hepatology* 2001;33:464–470. <https://doi.org/10.1053/jhep.2001.22172>.
- [16] Kim WR, Biggins SW, Kremers WK, Wiesner RH, Kamath PS, Benson JT, et al. Hyponatremia and mortality among patients on the liver-transplant waiting list. *N Engl J Med* 2008;359:1018–1026. <https://doi.org/10.1056/NEJMoa0801209>.
- [17] Kim WR, Mannalithara A, Heimbach JK, Kamath PS, Asrani SK, Biggins SW, et al. MELD 3.0: the model for end-stage liver disease updated for the modern era. *Gastroenterology* 2021;161:1887–1895.e4. <https://doi.org/10.1053/j.gastro.2021.08.050>.
- [18] Chong Lugon N, Rabiee A, Mezzacappa C, Kaplan DE, Taddei TH, Garcia-Tsao G. Prognostic significance of mild ascites in patients with cirrhosis. *Hepatology* 2025. <https://doi.org/10.1097/HEP.0000000000001452>.
- [19] Tonon M, Gagliardi R, Pompili E, Barone A, Zaccherini G, Zilio G, et al. Validation and expansion of Baveno VII recompensation criteria in patients with cirrhosis and curable liver disease. *J Hepatol* 2025;83:888–898. <https://doi.org/10.1016/j.jhep.2025.04.018>.
- [20] Sørensen J, Klee M, Palshof T, Hansen H. Performance status assessment in cancer patients. An inter-observer variability study. *Br J Cancer* 1993;67:773–775. <https://doi.org/10.1038/bjc.1993.140>.
- [21] Duvoux C, Roudot-Thoraval F, Decaens T, Pessione F, Badran H, Piardi T, et al. Liver transplantation for hepatocellular carcinoma: a model including α -fetoprotein improves the performance of milan criteria. *Gastroenterology* 2012;143:986–994.e3. <https://doi.org/10.1053/j.gastro.2012.05.052>.
- [22] Mazzaferro V, Sposito C, Zhou J, Pinna AD, De Carlis L, Fan J, et al. Metroticket 2.0 model for analysis of competing risks of death after liver transplantation for hepatocellular carcinoma. *Gastroenterology* 2018;154:128–139. <https://doi.org/10.1053/j.gastro.2017.09.025>.
- [23] Singal AG, Hoshida Y, Pinato DJ, Marrero J, Nault J-C, Paradis V, et al. International liver cancer association (ILCA) white paper on biomarker development for hepatocellular carcinoma. *Gastroenterology* 2021;160:2572–2584. <https://doi.org/10.1053/j.gastro.2021.01.233>.
- [24] Sieghart W, Pinter M, Huckle F, Graziadei I, Schöniger-Hekele M, Müller C, et al. Single determination of C-reactive protein at the time of diagnosis predicts long-term outcome of patients with hepatocellular carcinoma. *Hepatology* 2013;57:2224–2234. <https://doi.org/10.1002/hep.26057>.
- [25] Londoño M-C, Reig M. Multidisciplinary clinical approach to cancer patients with immune-related adverse events induced by checkpoint inhibitors. *Cancers (Basel)* 2020;12:3446. <https://doi.org/10.3390/cancers12113446>.
- [26] Bruix J, Reig M, Rimola J, Forner A, Burrel M, Vilana R, et al. Clinical decision making and research in hepatocellular carcinoma: pivotal role of imaging techniques. *Hepatology* 2010;54:2238–2244. <https://doi.org/10.1002/hep.24670>.
- [27] Matute-González, Mario; Ronot, Maxime; Chernyak, Victoria; Sangro, Bruno; Rimola, Jordi. Response evaluation to systemic therapy in HCC: Current challenges and future perspectives. *Information Hepatol*. <https://doi.org/10.1097/HEP.0000000000001621>. November 19, 2025.
- [28] Reig M, Rimola J, Torres F, Darnell A, Rodríguez-Lope C, Forner A, et al. Postprogression survival of patients with advanced hepatocellular carcinoma: rationale for second-line trial design. *Hepatology* 2013;58:2023–2031. <https://doi.org/10.1002/hep.26586>.
- [29] Torres F, Ríos J, Saez-Péñataro J, Pontes C. Is propensity score analysis a valid surrogate of randomization for the avoidance of allocation bias? *Semin Liver Dis* 2017;37:275–286. <https://doi.org/10.1055/s-0037-1606213>.
- [30] Kim DW. Statistical methods for baseline adjustment and cohort analysis in Korean national health insurance claims data: a review of PSM, IPTW, and survival analysis with future directions. *J Korean Med Sci* 2025;40. <https://doi.org/10.3346/jkms.2025.40.e110>.
- [31] Chattopadhyay A, Hase CH, Zubizarreta JR. Balancing vs modeling approaches to weighting in practice. *Stat Med* 2020;39:3227–3254. <https://doi.org/10.1002/sim.8659>.
- [32] Mishel MH. Reconceptualization of the uncertainty in illness theory. *Image J Nurs Scholarsh* 1990;22:256–262. <https://doi.org/10.1111/j.1547-5069.1990.tb00225.x>.
- [33] McCormick KM. A concept analysis of uncertainty in illness. *J Nurs Scholarsh* 2002;34:127–131. <https://doi.org/10.1111/j.1547-5069.2002.00127.x>.
- [34] Griffiths F, Green E, Tsouroufli M. The nature of medical evidence and its inherent uncertainty for the clinical consultation: qualitative study. *BMJ* 2005;330:511. <https://doi.org/10.1136/bmj.38336.482720.8F>.
- [35] Beumer BR, Polak WG, de Man RA, Metselaar HJ, van Klaveren D, Labrecque J, et al. Impact of waiting time on post-transplant survival for recipients with hepatocellular carcinoma: a natural experiment randomized by blood group. *JHEP Rep* 2023;5:100629. <https://doi.org/10.1016/j.jhepr.2022.100629>.
- [36] Ong DY, Lee ZY, Pua U. Impact of waiting time on hepatocellular carcinoma progression in patients undergoing curative tumour ablation. *Quant Imaging Med Surg* 2022;12:1499–1504. <https://doi.org/10.21037/qims-20-1411>.
- [37] Firl DJ, Kimura S, McVey J, Hashimoto K, Yeh H, Miller CM, et al. Reframing the approach to patients with hepatocellular carcinoma: longitudinal assessment with hazard associated with liver transplantation for HCC (HALTHCC) improves ablate and wait strategy. *Hepatology* 2018;68:1448–1458. <https://doi.org/10.1002/hep.29907>.
- [38] Mehta N, Sarkar M, Dodge JL, Fidelman N, Roberts JP, Yao FY. Intention to treat outcome of T1 hepatocellular carcinoma with the “wait and not ablate” approach until meeting T2 criteria for liver transplant listing. *Liver Transpl* 2016;22:178–187. <https://doi.org/10.1002/lt.24360>.
- [39] Marcantei C, Couret A, King J, Mazeaud S, Armand A, Ennequin G. Effects of exercise training on muscle mass and physical function in patients with hepatocellular carcinoma after diagnosis: a systematic review. *Dig Dis Sci* 2018;63:2667–2680. <https://doi.org/10.1007/s10620-024-08441-6>.
- [40] Vogel A, Cervantes A, Chau I, Daniele B, Llovet J, Meyer T, et al. Hepatocellular carcinoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Ann Oncol* 2018;29:iv238–iv255. <https://doi.org/10.1093/annonc/ndy308>.
- [41] Castelo-Branco L, Pellat A, Martins-Branco D, Valachis A, Derksen JW, Suijkerbuijk KPM, et al. ESMO guidance for reporting oncology real-world evidence (GROW). *Ann Oncol* 2023;34:1097–1112. <https://doi.org/10.1016/j.annonc.2023.10.001>.
- [42] Margini C, Berzigotti A. Portal vein thrombosis: the role of imaging in the clinical setting. *Dig Liver Dis* 2017;49:113–120. <https://doi.org/10.1016/j.dld.2016.11.013>.
- [43] Vilana R, Bru C, Bruix J, Castells A, Sole M, Rodes J. Fine-needle aspiration biopsy of portal vein thrombus: value in detecting malignant thrombosis. *Am J Roentgenol* 1993;160:1285–1287. <https://doi.org/10.2214/ajr.160.6.8388621>.
- [44] de Franco R, Bosch J, Garcia-Tsao G, Reiberger T, Ripoll C, Abraldes JG, et al. Baveno VII – renewing consensus in portal hypertension. *J Hepatol* 2022;76:959–974. <https://doi.org/10.1016/j.jhep.2021.12.022>.
- [45] Adachi E, Maeda T, Matsumata T, Shirabe K, Kinukawa N, Sugimachi K, et al. Risk factors for intrahepatic recurrence in human small hepatocellular carcinoma. *Gastroenterology* 1995;108:768–775. [https://doi.org/10.1016/0016-5085\(95\)90450-6](https://doi.org/10.1016/0016-5085(95)90450-6).
- [46] Imamura H, Matsuyama Y, Tanaka E, Ohkubo T, Hasegawa K, Miyagawa S, et al. Risk factors contributing to early and late phase intrahepatic recurrence of hepatocellular carcinoma after hepatectomy. *J Hepatol* 2003;38:200–207. [https://doi.org/10.1016/S0168-8278\(02\)00360-4](https://doi.org/10.1016/S0168-8278(02)00360-4).
- [47] Fuster-Anglada C, Mauro E, Ferrer-Fàbrega J, Caballol B, Sanduzzi-Zamparelli M, Bruix J, et al. Histological predictors of aggressive recurrence of hepatocellular carcinoma after liver resection. *J Hepatol* 2024;81:995–1004. <https://doi.org/10.1016/j.jhep.2024.06.018>.
- [48] Nagasue N, Uchida M, Makino Y, Takemoto Y, Yamanoi A, Hayashi T, et al. Incidence and factors associated with intrahepatic recurrence following resection of hepatocellular carcinoma. *Gastroenterology* 1993;105:488–494. [https://doi.org/10.1016/0016-5085\(93\)90724-Q](https://doi.org/10.1016/0016-5085(93)90724-Q).
- [49] Poon R-T, Fan S-T. Is primary resection and salvage transplantation for hepatocellular carcinoma a reasonable strategy? *Ann Surg* 2004;240:928. <https://doi.org/10.1097/01.sla.0000143808.63039.51>.

- [50] Shirabe K, Kanematsu T, Matsumata T, Adachi E, Akazawa K, Sugimachi K. Factors linked to early recurrence of small hepatocellular carcinoma after hepatectomy: univariate and multivariate analyses. *Hepatology* 1991;14:802–805. <https://doi.org/10.1002/hep.1840140510>.
- [51] Sala M, Fuster J, Llovet JM, Navasa M, Solé M, Varela M, et al. High pathological risk of recurrence after surgical resection for hepatocellular carcinoma: an indication for salvage liver transplantation. *Liver Transpl* 2004;10:1294–1300. <https://doi.org/10.1002/lt.20202>.
- [52] Ferrer-Fàbrega J, Forner A, Llicioni A, Miquel R, Molina V, Navasa M, et al. Prospective validation of ab initio liver transplantation in hepatocellular carcinoma upon detection of risk factors for recurrence after resection. *Hepatology* 2016;63:839–849. <https://doi.org/10.1002/hep.28339>.
- [53] Boros C, Sutter O, Cauchy F, Ganne-Carrié N, Nahon P, N'kontchou G, et al. Upfront multi-bipolar radiofrequency ablation for <sc>HCC</sc> in transplant-eligible cirrhotic patients with salvage transplantation in case of recurrence. *Liver Int* 2024;44:1464–1473. <https://doi.org/10.1111/liv.15900>.
- [54] Roberts JP, Venook A, Kerlan R, Yao F. Hepatocellular carcinoma: ablate and wait versus rapid transplantation. *Liver Transpl* 2010;16:925–929. <https://doi.org/10.1002/lt.22103>.
- [55] Takayama T, Hasegawa K, Izumi N, Kudo M, Shimada M, Yamanaka N, et al. Surgery versus radiofrequency ablation for small hepatocellular carcinoma: a randomized controlled trial (SURF trial). *Liver Cancer* 2022;11:209–218. <https://doi.org/10.1159/000521665>.
- [56] Cucchetti A, Trevisani F, Pecorelli A, Erroi V, Farinati F, Ciccarese F, et al. Estimation of lead-time bias and its impact on the outcome of surveillance for the early diagnosis of hepatocellular carcinoma. *J Hepatol* 2014;61:333–341. <https://doi.org/10.1016/j.jhep.2014.03.037>.
- [57] Feng K, Yan J, Li X, Xia F, Ma K, Wang S, et al. A randomized controlled trial of radiofrequency ablation and surgical resection in the treatment of small hepatocellular carcinoma. *J Hepatol* 2012;57:794–802. <https://doi.org/10.1016/j.jhep.2012.05.007>.
- [58] Cho YK, Kim JK, Kim WT, Chung JW. Hepatic resection versus radiofrequency ablation for very early stage hepatocellular carcinoma. *Hepatology* 2010;51:1284–1290. <https://doi.org/10.1002/hep.23466>.
- [59] Doyle A, Gorgen A, Muaddi H, Aravinthan AD, Issachar A, Mironov O, et al. Outcomes of radiofrequency ablation as first-line therapy for hepatocellular carcinoma less than 3 cm in potentially transplantable patients. *J Hepatol* 2019;70:866–873. <https://doi.org/10.1016/j.jhep.2018.12.027>.
- [60] Giulianti F, Famularo S, Grasselli S, Sangiovanni A, Vitale A, Cabibbo G, et al. Minimally invasive hepatectomy vs. thermoablation for single small (<3 cm) hepatocellular carcinoma: a weighted real-life national comparison. *JHEP Rep* 2025;7:101420. <https://doi.org/10.1016/j.jhepr.2025.101420>.
- [61] Li Z, Pfister M, Raptis DA, Ma Y, Yang G, Li L, et al. Novel benchmark for robotic liver resection – bridging tradition with innovation. *Ann Surg* 2025. <https://doi.org/10.1097/SLA.0000000000006759>.
- [62] Kabir T, Tan ZZ, Syn NL, Wu E, Lin JD, Zhao JJ, et al. Laparoscopic versus open resection of hepatocellular carcinoma in patients with cirrhosis: meta-analysis. *Br J Surg* 2021;109:21–29. <https://doi.org/10.1093/bjs/znab376>.
- [63] Kim E, Sher A, Abboud G, Schwartz M, Facciuto M, Tabrizian P, et al. Radiation segmentectomy for curative intent of unresectable very early to early stage hepatocellular carcinoma (RASER): a single-centre, single-arm study. *Lancet Gastroenterol Hepatol* 2022;7:843–850. [https://doi.org/10.1016/S2468-1253\(22\)00091-7](https://doi.org/10.1016/S2468-1253(22)00091-7).
- [64] Sugimoto K, Imajo K, Kuroda H, Murohisa G, Shiozawa K, Sakamaki K, et al. Microwave ablation vs. single-needle radiofrequency ablation for the treatment of HCC up to 4 cm: a randomized-controlled trial. *JHEP Rep* 2025;7:101269. <https://doi.org/10.1016/j.jhepr.2024.101269>.
- [65] Salem R, Johnson GE, Kim E, Riaz A, Bishay V, Boucher E, et al. Yttrium-90 radioembolization for the treatment of solitary, unresectable HCC: the LEGACY study. *Hepatology* 2021;74:2342–2352. <https://doi.org/10.1002/HEP.31819>.
- [66] Xi M, Yang Z, Hu L, Fu Y, Hu D, Zhou Z, et al. Radiofrequency ablation versus stereotactic body radiotherapy for recurrent small hepatocellular carcinoma: a randomized, open-label, controlled trial. *J Clin Oncol* 2025;43:1073–1082. <https://doi.org/10.1200/JCO-24-01532>.
- [67] Bruix J, Castells A, Bosch J, Feu F, Fuster J, Garcia-Pagan J, et al. Surgical resection of hepatocellular carcinoma in cirrhotic patients: prognostic value of preoperative portal pressure. *Gastroenterology* 1996;111:1018–1022. [https://doi.org/10.1016/S0016-5085\(96\)70070-7](https://doi.org/10.1016/S0016-5085(96)70070-7).
- [68] Berzigotti A, Reig M, Abraldes JG, Bosch J, Bruix J. Portal hypertension and the outcome of surgery for hepatocellular carcinoma in compensated cirrhosis: a systematic review and meta-analysis. *Hepatology* 2015;61:526–536. <https://doi.org/10.1002/hep.27431>.
- [69] Citterio D, Facciorusso A, Sposito C, Rota R, Bhoori S, Mazzaferro V. Hierarchic interaction of factors associated with liver decompensation after resection for hepatocellular carcinoma. *JAMA Surg* 2016;151:846. <https://doi.org/10.1001/jamasurg.2016.1121>.
- [70] European Association for the Study of the Liver. EASL Clinical Practice Guidelines on non-invasive tests for evaluation of liver disease severity and prognosis – 2021 update. *J Hepatol* 2021;75:659–689. <https://doi.org/10.1016/j.jhep.2021.05.025>.
- [71] Ferrusquia-Acosta J, Bassegoda O, Turco L, Reverter E, Pellone M, Bianchini M, et al. Agreement between wedged hepatic venous pressure and portal pressure in non-alcoholic steatohepatitis-related cirrhosis. *J Hepatol* 2021;74:811–818. <https://doi.org/10.1016/j.jhep.2020.10.003>.
- [72] Baffy G, Bosch J. Overlooked subclinical portal hypertension in non-cirrhotic NAFLD: is it real and how to measure it? *J Hepatol* 2022;76:458–463. <https://doi.org/10.1016/j.jhep.2021.09.029>.
- [73] Lens S, Baiges A, Alvarado-Tapias E, Llop E, Martínez J, Fortea JL, et al. Clinical outcome and hemodynamic changes following HCV eradication with oral antiviral therapy in patients with clinically significant portal hypertension. *J Hepatol* 2020;73:1415–1424. <https://doi.org/10.1016/j.jhep.2020.05.050>.
- [74] Fang J, Zhou H, Zhang C, Shang L, Zhang L, Xu J, et al. A novel vascular pattern promotes metastasis of hepatocellular carcinoma in an epithelial-mesenchymal transition-independent manner. *Hepatology* 2015;62:452–465. <https://doi.org/10.1002/hep.27760>.
- [75] Leirich BM, Zhang J, Monga SP, Dhanasekaran R. Battle of the biopsies: role of tissue and liquid biopsy in hepatocellular carcinoma. *J Hepatol* 2024;80:515–530. <https://doi.org/10.1016/j.jhep.2023.11.030>.
- [76] Campani C, Zucman-Rossi J, Nault J-C. Genetics of hepatocellular carcinoma: from tumor to circulating DNA. *Cancers (Basel)* 2023;15:817. <https://doi.org/10.3390/cancers15030817>.
- [77] Berardi G, Morise Z, Sposito C, Igarashi K, Panetta V, Simonelli I, et al. Development of a nomogram to predict outcome after liver resection for hepatocellular carcinoma in Child-Pugh B cirrhosis. *J Hepatol* 2020;72:75–84. <https://doi.org/10.1016/j.jhep.2019.08.032>.
- [78] Molina V, Sampson-Dávila J, Ferrer J, Fondevila C, Díaz del Gobbo R, Calatayud D, et al. Benefits of laparoscopic liver resection in patients with hepatocellular carcinoma and portal hypertension: a case-matched study. *Surg Endosc* 2018;32:2345–2354. <https://doi.org/10.1007/s00464-017-5930-1>.
- [79] Mazzaferro V, Regalia E, Doci R, Andreola S, Pulvirenti A, Bozzetti F, et al. Liver transplantation for the treatment of small hepatocellular carcinomas in patients with cirrhosis. *N Engl J Med* 1996;334:693–700. <https://doi.org/10.1056/NEJM199603143341104>.
- [80] Ishizawa T, Hasegawa K, Aoki T, Takahashi M, Inoue Y, Sano K, et al. Neither multiple tumors nor portal hypertension are surgical contraindications for hepatocellular carcinoma. *Gastroenterology* 2008;134:1908–1916. <https://doi.org/10.1053/j.gastro.2008.02.091>.
- [81] Shiina S, Tateishi R, Arano T, Uchino K, Enoku K, Nakagawa H, et al. Radiofrequency ablation for hepatocellular carcinoma: 10-year outcome and prognostic factors. *Am J Gastroenterol* 2012;107:569–577. <https://doi.org/10.1038/ajg.2011.425>.
- [82] Shiina S, Teratani T, Obi S, Sato S, Tateishi R, Fujishima T, et al. A randomized controlled trial of radiofrequency ablation with ethanol injection for small hepatocellular carcinoma. *Gastroenterology* 2005;129:122–130. <https://doi.org/10.1053/j.gastro.2005.04.009>.
- [83] Germani G, Pleguezuelo M, Gurusamy K, Meyer T, Isgró G, Burroughs AK. Clinical outcomes of radiofrequency ablation, percutaneous alcohol and acetic acid injection for hepatocellular carcinoma: a meta-analysis. *J Hepatol* 2010;52:380–388. <https://doi.org/10.1016/j.jhep.2009.12.004>.
- [84] Méndez Romero A, van der Holt B, Willemssen FEJA, de Man RA, Heijmen BJM, Habraken S, et al. Transarterial chemoembolization with drug-eluting beads versus stereotactic body radiation therapy for hepatocellular carcinoma: outcomes from a multicenter, randomized, phase 2 trial (the TRENDY trial). *Int J Radiat Oncol* 2023;117:45–52. <https://doi.org/10.1016/j.ijrobp.2023.03.064>.
- [85] Takeda A, Sanuki N, Tsurugai Y, Iwabuchi S, Matsunaga K, Ebinuma H, et al. Phase 2 study of stereotactic body radiotherapy and optional transarterial chemoembolization for solitary hepatocellular carcinoma not amenable to resection and radiofrequency ablation. *Cancer* 2016;122:2041–2049. <https://doi.org/10.1002/cncr.30008>.
- [86] Sapichochin G, Barry A, Doherty M, Fischer S, Goldaracena N, Rosales R, et al. Stereotactic body radiotherapy vs. TACE or RFA as a bridge to

- transplant in patients with hepatocellular carcinoma. An intention-to-treat analysis. *J Hepatol* 2017;67:92–99. <https://doi.org/10.1016/j.jhep.2017.02.022>.
- [87] Moon MA, Yanagihara TK, Dawson LAJ, Il Yu J, Lawrence T, Kim TH, et al. Overall survival among patients with hepatocellular carcinoma treated with external beam radiation therapy: individual patient data outcomes from a multinational cohort. *ILCA* 2025.
- [88] Salem R, Padia SA, Lam M, Chiesa C, Haste P, Sangro B, et al. Clinical, dosimetric, and reporting considerations for Y-90 glass microspheres in hepatocellular carcinoma: updated 2022 recommendations from an international multidisciplinary working group. *Eur J Nucl Med Mol Imaging* 2023;50:328–343. <https://doi.org/10.1007/s00259-022-05956-w>.
- [89] Lewandowski RJ, Gabr A, Abouchaleh N, Ali R, Al Asadi A, Mora RA, et al. Radiation segmentectomy: potential curative therapy for early hepatocellular carcinoma. *Radiology* 2018;287:1050–1058. <https://doi.org/10.1148/radiol.2018171768>.
- [90] Dhondt E, Lambert B, Hermie L, Huyck L, Vanlangenhove P, Geerts A, et al. 90 Y radioembolization versus drug-eluting bead chemoembolization for unresectable hepatocellular carcinoma: results from the TRACE phase II randomized controlled trial. *Radiology* 2022;303:699–710. <https://doi.org/10.1148/radiol.211806>.
- [91] Vouche M, Lewandowski RJ, Atassi R, Memon K, Gates VL, Ryu RK, et al. Radiation lobectomy: time-dependent analysis of future liver remnant volume in unresectable liver cancer as a bridge to resection. *J Hepatol* 2013;59:1029–1036. <https://doi.org/10.1016/j.jhep.2013.06.015>.
- [92] Yanagihara TK, Tepper JE, Moon AM, Barry A, Molla M, Seong J, et al. Defining minimum treatment parameters of ablative radiation therapy in patients with hepatocellular carcinoma: an expert consensus. *Pract Radiat Oncol* 2024;14:134–145. <https://doi.org/10.1016/j.prro.2023.08.016>.
- [93] Jang W Il, Jo S, Moon JE, Bae SH, Park HC. The current evidence of intensity-modulated radiotherapy for hepatocellular carcinoma: a systematic review and meta-analysis. *Cancers (Basel)* 2023;15:4914. <https://doi.org/10.3390/cancers15204914>.
- [94] Bush DA, Volk M, Smith JC, Reeves ME, Sanghvi S, Slater JD, et al. Proton beam radiotherapy versus transarterial chemoembolization for hepatocellular carcinoma: results of a randomized clinical trial. *Cancer* 2023;129:3554–3563. <https://doi.org/10.1002/cncr.34965>.
- [95] Comito T, Loi M, Franzese C, Clerici E, Franceschini D, Badalamenti M, et al. Stereotactic radiotherapy after incomplete transarterial (Chemo-) embolization (TAE/TACE) versus exclusive TAE or TACE for treatment of inoperable HCC: a phase III trial (NCT02323360). *Curr Oncol* 2022;29:8802–8813. <https://doi.org/10.3390/curroncol29110692>.
- [96] Dharancy S, Boitard J, Decaens T, Sergent G, Boleslawski E, Duvoux C, et al. Comparison of two techniques of transarterial chemoembolization before liver transplantation for hepatocellular carcinoma: a case-control study. *Liver Transpl* 2007;13:665–671. <https://doi.org/10.1002/lt.21109>.
- [97] Agopian VG, Morshedi MM, McWilliams J, Harlander-Locke MP, Markovic D, Zarrinpar A, et al. Complete pathologic response to pre-transplant locoregional therapy for hepatocellular carcinoma defines cancer cure after liver transplantation. *Ann Surg* 2015;262:536–545. <https://doi.org/10.1097/SLA.0000000000001384>.
- [98] Allard M-A, Sebahg M, Ruiz A, Guettier C, Paule B, Vibert E, et al. Does pathological response after transarterial chemoembolization for hepatocellular carcinoma in cirrhotic patients with cirrhosis predict outcome after liver resection or transplantation? *J Hepatol* 2015;63:83–92. <https://doi.org/10.1016/j.jhep.2015.01.023>.
- [99] Harnois DM, Steers J, Andrews JC, Rubin JC, Pitot HC, Burgart L, et al. Preoperative hepatic artery chemoembolization followed by orthotopic liver transplantation for hepatocellular carcinoma. *Liver Transpl Surg* 1999;5:192–199. <https://doi.org/10.1002/lt.500050307>.
- [100] Golfieri R, Cappelli A, Cucchetti A, Piscaglia F, Carpenzano M, Peri E, et al. Efficacy of selective transarterial chemoembolization in inducing tumor necrosis in small (<5 cm) hepatocellular carcinomas. *Hepatology* 2011;53:1580–1589. <https://doi.org/10.1002/hep.24246>.
- [101] Badar W, Cooper EG, Florido CR, Rabaza M, Sheikh U, Guzman G, et al. Comparative radiologic response assessment after transarterial chemoembolization, percutaneous ablation, and multimodal treatment: radiologic-pathologic correlation in 81 tumors. *J Vasc Interv Radiol* 2025;36:961–970. <https://doi.org/10.1016/j.jvir.2025.02.016>.
- [102] Kim NG, Yao FY, Kwong AJ, Mehta N. Yttrium-90 radioembolization for hepatocellular carcinoma is associated with a lower risk of liver transplant waitlist dropout compared to chemoembolization. *J Hepatol* 2026. <https://doi.org/10.1016/j.jhep.2026.01.008>.
- [103] Li Z, Yan M, Claassen MPAW, Rajendran L, Magyar CTJ, Saha S, et al. SBRT for bridging and downstaging HCC before transplant. *JHEP Rep* 2025;7:101451. <https://doi.org/10.1016/j.jhep.2025.101451>.
- [104] Cathomas M, Mueller F, Mertineit N, Baumgartner I, Candinas D, Berzigotti A, et al. Comparison of transarterial bland embolization and drug-eluting beads transarterial chemoembolization for very early and early hepatocellular carcinoma not amenable for surgery or ablation: a single center retrospective data analysis. *J Gastrointest Oncol* 2023;14:2167–2177. <https://doi.org/10.21037/jgo-23-261>.
- [105] Hashem E, Sait S, Thomas DN, Watson C, Moeen S, Peddu P. Transarterial chemoembolisation for very early and early stage hepatocellular carcinoma: single-centre experience. *Clin Radiol* 2023;78:e113–e122. <https://doi.org/10.1016/j.crad.2022.09.120>.
- [106] Sher A, Todd RI, Shilo D, Garcia-Reyes KI, Bishay VL, Patel RI, et al. Histopathologic response and oncologic outcomes after segmental and subsegmental transarterial chemoembolization and radioembolization for hepatocellular carcinoma. *J Vasc Interv Radiol* 2025;36:1285–1295.e2. <https://doi.org/10.1016/j.jvir.2025.05.009>.
- [107] Salem R, Padia SA, Toskich BB, Callahan JD, Fowers KD, Geller BS, et al. Radiation segmentectomy for early hepatocellular carcinoma is curative. *J Hepatol* 2025;82:1125–1132. <https://doi.org/10.1016/j.jhep.2025.01.005>.
- [108] Mehta N, Frenette C, Tabrizian P, Hoteit M, Guy J, Parikh N, et al. Downstaging outcomes for hepatocellular carcinoma: results from the multicenter evaluation of reduction in tumor size before liver transplantation (MERITS-LT) consortium. *Gastroenterology* 2021;161:1502–1512. <https://doi.org/10.1053/j.gastro.2021.07.033>.
- [109] Vitale A, Romano P, Cillo U, Busti M, Nezi G, De Carlis R, et al. Liver resection vs nonsurgical treatments for patients with early multinodular hepatocellular carcinoma. *JAMA Surg* 2024;159:881. <https://doi.org/10.1001/jamasurg.2024.1184>.
- [110] Yu S, Jo H-S, Yu Y-D, Choi Y jin, Kim D-S. Clinical outcome of surgical resection for multifocal T2-T3 hepatocellular carcinoma up to 3 nodules: a comparative analysis with a single nodule. *J Liver Cancer* 2023;23:377–388. <https://doi.org/10.17998/jlc.2023.08.24>.
- [111] Zhang Y-J, Chen J, Zhou Z, Hu D, Wang J, Pan Y, et al. Transarterial chemoembolization with radiofrequency ablation versus surgical resection for small late-recurrence hepatocellular carcinoma. *Radiology* 2025;314. <https://doi.org/10.1148/radiol.241096>.
- [112] Yuan Y, Peng H, He W, Zheng Y, Qiu J, Chen B, et al. Partial hepatectomy versus interventional treatment in patients with hepatitis B virus-related hepatocellular carcinoma and clinically significant portal hypertension: a randomized comparative clinical trial. *Cancer Commun* 2024;44:1337–1349. <https://doi.org/10.1002/cac2.12614>.
- [113] Fang C, Luo R, Zhang Y, Wang J, Feng K, Liu S, et al. Hepatectomy versus transcatheter arterial chemoembolization for resectable BCLC stage A/B hepatocellular carcinoma beyond Milan criteria: a randomized clinical trial. *Front Oncol* 2023;13. <https://doi.org/10.3389/fonc.2023.1101162>.
- [114] Wang Z, Zhang H, Meng Q, Zhang D, Wu S, Hong Z, et al. A multicenter case-controlled study on laparoscopic hepatectomy versus microwave ablation as first-line therapy for 3–5 cm hepatocellular carcinoma in patients aged 60 and older. *Int J Surg* 2024;110:1356–1366. <https://doi.org/10.1097/JS9.0000000000000839>.
- [115] Kawaguchi Y, Hasegawa K, Kashiwabara K, Okamura Y, Kurosaki M, Kudo M, et al. Surgery versus ablation for hepatocellular carcinoma: a randomized controlled trial (SURF-RCT trial) and a nonrandomized prospective observational trial (SURF-Cohort trial). *J Clin Oncol* 2025;43:2628–2638. <https://doi.org/10.1200/JCO-24-02030>.
- [116] Qin S, Chen M, Cheng A-L, Kaseb AO, Kudo M, Lee HC, et al. Atezolizumab plus bevacizumab versus active surveillance in patients with resected or ablated high-risk hepatocellular carcinoma (IMbrave050): a randomised, open-label, multicentre, phase 3 trial. *Lancet* 2023;402:1835–1847. [https://doi.org/10.1016/S0140-6736\(23\)01796-8](https://doi.org/10.1016/S0140-6736(23)01796-8).
- [117] Fröhling P, Nilsson A, Duraj F, Haglund U, Norén A. Single-center non-randomized clinical trial to assess the safety and efficacy of irreversible electroporation (IRE) ablation of liver tumors in humans: short to mid-term results. *Eur J Surg Oncol* 2017;43:751–757. <https://doi.org/10.1016/j.ejso.2016.12.004>.
- [118] Rog CJ, Alassas M, Ong E. Histotripsy—a novel and intriguing technique of liver ablation. *JAMA Surg* 2025;160:100. <https://doi.org/10.1001/jamasurg.2024.2339>.
- [119] Wehrle CJ, Burns K, Ong E, Couillard A, Parikh ND, Caoili E, et al. The first international experience with histotripsy: a safety analysis of 230 cases.

- J Gastrointest Surg 2025;29:102000. <https://doi.org/10.1016/j.gassur.2025.102000>.
- [120] Mendiratta-Lala M, Wiggermann P, Pech M, Serres-Créixams X, White SB, Davis C, et al. The #HOPE4LIVER single-arm pivotal trial for histotripsy of primary and metastatic liver tumors. *Radiology* 2024;312. <https://doi.org/10.1148/radiol.233051>.
- [121] Lee B, Han H-S, Yoon Y-S, Cho JY, Lee HW, Lee J-H, et al. Treatment strategies for solitary hepatocellular carcinoma: comparative outcomes of radiofrequency ablation vs. laparoscopic liver resection based on tumor location. *Surg Endosc* 2025;39:2175–2184. <https://doi.org/10.1007/s00464-025-11566-y>.
- [122] Wei Y, Zhang L, Zhang S, Song M, Ji C. Laparoscopic-assisted microwave ablation in treatment of small hepatocellular carcinoma: safety and efficacy in comparison with laparoscopic hepatectomy. *BMC Surg* 2024;24:138. <https://doi.org/10.1186/s12893-024-02432-0>.
- [123] Yu H, Zhao F, Men X, Zhu H, Yan J, Liu Z, et al. Microwave ablation versus laparoscopic resection for hepatocellular carcinoma in patients with clinically significant portal hypertension: a propensity score-matched study of postoperative liver decompensation. *Eur Radiol* 2023;34:3226–3235. <https://doi.org/10.1007/s00330-023-10268-0>.
- [124] Muñoz-Martínez S, Sapena V, García-Criado Á, Darnell A, Forner A, Belmonte E, et al. Risk of treatment failure and death after ablation in hepatocellular carcinoma patients—a multiparametric prediction. *Cancers (Basel)* 2023;15:3269. <https://doi.org/10.3390/cancers15133269>.
- [125] Yang Y, Chen Y, Zhang X, Xin Y, Wang Y, Li X, et al. Predictors and patterns of recurrence after radiofrequency ablation for hepatocellular carcinoma within up-to-seven criteria: a multicenter retrospective study. *Eur J Radiol* 2021;138:109623. <https://doi.org/10.1016/j.ejrad.2021.109623>.
- [126] Hori T, Nagata K, Hasuiki S, Onaga M, Motoda M, Moriuchi A, et al. Risk factors for the local recurrence of hepatocellular carcinoma after a single session of percutaneous radiofrequency ablation. *J Gastroenterol* 2003;38:977–981. <https://doi.org/10.1007/s00535-003-1181-0>.
- [127] Yang B, Zou J, Xia J, Ren Z, Gan Y, Wang Y, et al. Risk factors for recurrence of small hepatocellular carcinoma after long-term follow-up of percutaneous radiofrequency ablation. *Eur J Radiol* 2011;79:196–200. <https://doi.org/10.1016/j.ejrad.2010.02.010>.
- [128] Komorizono Y, Oketani M, Sako K, Yamasaki N, Shibata T, Maeda M, et al. Risk factors for local recurrence of small hepatocellular carcinoma tumors after a single session, single application of percutaneous radiofrequency ablation. *Cancer* 2003;97:1253–1262. <https://doi.org/10.1002/cncr.11168>.
- [129] Zytoon AA, Ishii H, Murakami K, El-Kholi MR, Furuse J, El-Dorri A, et al. Recurrence-free survival after radiofrequency ablation of hepatocellular carcinoma. A registry report of the impact of risk factors on outcome. *Jpn J Clin Oncol* 2007;37:658–672. <https://doi.org/10.1093/jco/hym086>.
- [130] Cho J-Y, Choi MS, Lee GS, Sohn W, Ahn J, Sinn D-H, et al. Clinical significance and predictive factors of early massive recurrence after radiofrequency ablation in patients with a single small hepatocellular carcinoma. *Clin Mol Hepatol* 2016;22:477–486. <https://doi.org/10.3350/cmh.2016.0048>.
- [131] Yang T, Liu D-Q, Qiu W, Fan Z-Q, Sun L-Y, Wang N-Y, et al. The Barthel Index predicts surgical textbook outcomes following hepatectomy for elderly patients with hepatocellular carcinoma: a multicenter cohort study. *Am J Surg* 2024;237:115761. <https://doi.org/10.1016/j.amjsurg.2024.05.002>.
- [132] Mehta N, Bhargui P, Yao FY, Mazzaferro V, Toso C, Akamatsu N, et al. Liver transplantation for hepatocellular carcinoma. Working group report from the ILTS transplant oncology consensus conference. *Transplantation* 2020;104:1136–1142. <https://doi.org/10.1097/TP.0000000000003174>.
- [133] Kodali S, Kulik L, D'Allesio A, De Martin E, Hakeem AR, Lewinska M, et al. The 2024 ILTS-ILCA consensus recommendations for liver transplantation for HCC and intrahepatic cholangiocarcinoma. *Liver Transpl* 2025. <https://doi.org/10.1097/LVT.0000000000000589>.
- [134] Bruix J, Sala M, Llovet JM. Chemoembolization for hepatocellular carcinoma. *Gastroenterology* 2004;127:S179–S188. <https://doi.org/10.1053/j.gastro.2004.09.032>.
- [135] Golfieri R, Giampalma E, Renzulli M, Cioni R, Bargellini I, Bartolozzi C, et al. Randomised controlled trial of doxorubicin-eluting beads vs conventional chemoembolisation for hepatocellular carcinoma. *Br J Cancer* 2014;111:255–264. <https://doi.org/10.1038/bjc.2014.199>.
- [136] Llovet JM, Real MI, Montaña X, Planas R, Coll S, Aponte J, et al. Arterial embolisation or chemoembolisation versus symptomatic treatment in patients with unresectable hepatocellular carcinoma: a randomised controlled trial. *Lancet* 2002;359:1734–1739. [https://doi.org/10.1016/S0140-6736\(02\)08649-X](https://doi.org/10.1016/S0140-6736(02)08649-X).
- [137] Guib B, Deschamps F, Aho S, Munck F, Dromain C, Boige V, et al. Liver/biliary injuries following chemoembolisation of endocrine tumours and hepatocellular carcinoma: lipiodol vs. drug-eluting beads. *J Hepatol* 2012;56:609–617. <https://doi.org/10.1016/j.jhep.2011.09.012>.
- [138] Monier A, Guib B, Duran R, Aho S, Bize P, Deltenre P, et al. Liver and biliary damages following transarterial chemoembolization of hepatocellular carcinoma: comparison between drug-eluting beads and lipiodol emulsion. *Eur Radiol* 2017;27:1431–1439. <https://doi.org/10.1007/s00330-016-4488-y>.
- [139] Malagari K, Pomoni M, Kelekis A, Pomoni A, Dourakis S, Spyridopoulos T, et al. Prospective randomized comparison of chemoembolization with doxorubicin-eluting beads and bland embolization with BeadBlock for hepatocellular carcinoma. *Cardiovasc Intervent Radiol* 2010;33:541–551. <https://doi.org/10.1007/s00270-009-9750-0>.
- [140] Brown KT, Do RK, Gonen M, Covey AM, Getrajdman GI, Sofocleous CT, et al. Randomized trial of hepatic artery embolization for hepatocellular carcinoma using doxorubicin-eluting microspheres compared with embolization with microspheres alone. *J Clin Oncol* 2016;34:2046–2053. <https://doi.org/10.1200/JCO.2015.64.0821>.
- [141] Zhang H, Xu J, Meng H, Shen L. Comparison of the effectiveness of transarterial bland embolization and transarterial chemoembolization for intermediate-stage hepatocellular carcinoma: a propensity score-matched study of 1,008 patients. *J Vasc Interv Radiol* 2025;36:41–49. <https://doi.org/10.1016/j.jvir.2024.09.005>.
- [142] Roth GS, Benhamou M, Teyssier Y, Seigneurin A, Abousalihac M, Sengel C, et al. Comparison of trans-arterial chemoembolization and bland embolization for the treatment of hepatocellular carcinoma: a propensity score analysis. *Cancers (Basel)* 2021;13:812. <https://doi.org/10.3390/cancers13040812>.
- [143] Zhu H-D, Li X, Sun J-H, Zhu X, Liu Z-Y, Li H-L, et al. Transarterial chemoembolization with epirubicin-loaded microspheres for hepatocellular carcinoma: a prospective, single-arm, multicenter, phase 2 study (STOPPER trial). *Cardiovasc Intervent Radiol* 2024;47:325–336. <https://doi.org/10.1007/s00270-024-03666-4>.
- [144] Wang Q, Xia D, Bai W, Wang E, Sun J, Huang M, et al. Development of a prognostic score for recommended TACE candidates with hepatocellular carcinoma: a multicentre observational study. *J Hepatol* 2019;70:893–903. <https://doi.org/10.1016/j.jhep.2019.01.013>.
- [145] Hameed B, Mehta N, Sapisochin G, Roberts JP, Yao FY. Alpha-fetoprotein level > 1000 ng/mL as an exclusion criterion for liver transplantation in patients with hepatocellular carcinoma meeting the Milan criteria. *Liver Transpl* 2014;20:945–951. <https://doi.org/10.1002/lt.23904>.
- [146] Yao FY, Mehta N, Flemming J, Dodge J, Hameed B, Fix O, et al. Downstaging of hepatocellular cancer before liver transplantation: long-term outcome compared to tumors within Milan criteria. *Hepatology* 2015;61:1968–1977. <https://doi.org/10.1002/hep.27752>.
- [147] Mazzaferro V, Citterio D, Bhoori S, Bongini M, Miceli R, De Carlis L, et al. Liver transplantation in hepatocellular carcinoma after tumour downstaging (XXL): a randomised, controlled, phase 2b/3 trial. *Lancet Oncol* 2020;21:947–956. [https://doi.org/10.1016/S1470-2045\(20\)30224-2](https://doi.org/10.1016/S1470-2045(20)30224-2).
- [148] Sapisochin G, Bruix J. Liver transplantation for hepatocellular carcinoma: outcomes and novel surgical approaches. *Nat Rev Gastroenterol Hepatol* 2017;14:203–217. <https://doi.org/10.1038/nrgastro.2016.193>.
- [149] Navasa M, Bruix J. Multifaceted perspective of the waiting list for liver transplantation: the value of pharmacokinetic models. *Hepatology* 2010;51:12–15. <https://doi.org/10.1002/hep.23332>.
- [150] Lammer J, Malagari K, Vogl T, Pilleul F, Denys A, Watkinson A, et al. Prospective randomized study of doxorubicin-eluting-bead embolization in the treatment of hepatocellular carcinoma: results of the PRECISION V study. *Cardiovasc Intervent Radiol* 2010;33:41–52. <https://doi.org/10.1007/s00270-009-9711-7>.
- [151] Burrel M, Reig M, Forner A, Barrufet M, Lope CR de, Tremosini S, et al. Survival of patients with hepatocellular carcinoma treated by transarterial chemoembolisation (TACE) using Drug Eluting Beads. Implications for clinical practice and trial design. *J Hepatol* 2012;56:1330–1335. <https://doi.org/10.1016/j.jhep.2012.01.008>.
- [152] Takayasu K, Arai S, Kudo M, Ichida T, Matsui O, Izumi N, et al. Super-selective transarterial chemoembolization for hepatocellular carcinoma. Validation of treatment algorithm proposed by Japanese guidelines. *J Hepatol* 2012;56:886–892. <https://doi.org/10.1016/j.jhep.2011.10.021>.
- [153] Veloso Gomes F, de Baère T, Verset G, Coimbra É, Tovar-Felice G, Malagari K, et al. Transarterial chemoembolization with anthracyclines-

- loaded polyethylene glycol drug eluting microspheres for the treatment of hepatocellular carcinoma: a pooled multicentric analysis of survival in 580 patients. *Cardiovasc Intervent Radiol* 2023;46:436–446. <https://doi.org/10.1007/s00270-023-03362-9>.
- [154] **Kudo M, Ueshima K, Ikeda M, Torimura T, Tanabe N, Aikata H, et al.** Final results of TACTICS: a randomized, prospective trial comparing transarterial chemoembolization plus sorafenib to transarterial chemoembolization alone in patients with unresectable hepatocellular carcinoma. *Liver Cancer* 2022;11:354–367. <https://doi.org/10.1159/000522547>.
- [155] Salem R, Lewandowski RJ, Kulik L, Wang E, Riaz A, Ryu RK, et al. Radioembolization results in longer time-to-progression and reduced toxicity compared with chemoembolization in patients with hepatocellular carcinoma. *Gastroenterology* 2011;140:497–507.e2. <https://doi.org/10.1053/j.gastro.2010.10.049>.
- [156] Salem R, Gordon AC, Mouli S, Hickey R, Kallini J, Gabr A, et al. Y90 radioembolization significantly prolongs time to progression compared with chemoembolization in patients with hepatocellular carcinoma. *Gastroenterology* 2016;151:1155–1163.e2. <https://doi.org/10.1053/j.gastro.2016.08.029>.
- [157] Carr BI, Kondragunta V, Buch SC, Branch RA. Therapeutic equivalence in survival for hepatic arterial chemoembolization and yttrium 90 microsphere treatments in unresectable hepatocellular carcinoma. *Cancer* 2010;116:1305–1314. <https://doi.org/10.1002/cncr.24884>.
- [158] Cabibbo G, Enea M, Attanasio M, Bruix J, Craxi A, Cammà C. A meta-analysis of survival rates of untreated patients in randomized clinical trials of hepatocellular carcinoma. *Hepatology* 2010;51:1274–1283. <https://doi.org/10.1002/hep.23485>.
- [159] Antkowiak M, Gabr A, Das A, Ali R, Kulik L, Ganger D, et al. Prognostic role of albumin, bilirubin, and ALBI scores: analysis of 1000 patients with hepatocellular carcinoma undergoing radioembolization. *Cancers (Basel)* 2019;11:879. <https://doi.org/10.3390/cancers11060879>.
- [160] Ranganathan S, Gabr A, Entezari P, Gordon AC, Riaz A, Desai K, et al. Radioembolization for intermediate-stage hepatocellular carcinoma maintains liver function and permits systemic therapy at progression. *J Vasc Interv Radiol* 2023;34:968–975. <https://doi.org/10.1016/j.jvir.2022.11.036>.
- [161] Kuehnl RA, Tcheleni LT, Ludmir EB, Escorcia FE, Shrestha S, Sanford N, et al. Meta-analysis of randomized controlled trials of external-beam radiation versus transarterial chemoembolization for hepatocellular carcinoma. *Cancer* 2025;131. <https://doi.org/10.1002/cncr.35720>.
- [162] Sorin Y, Ikeda K, Kawamura Y, Fujiyama S, Kobayashi M, Hosaka T, et al. Effectiveness of particle radiotherapy in various stages of hepatocellular carcinoma: a pilot study. *Liver Cancer* 2018;7:323–334. <https://doi.org/10.1159/000487311>.
- [163] **Garin E, Tselikas L, Guib B, Chalaye J, Edeline J, de Baere T, et al.** Personalised versus standard dosimetry approach of selective internal radiation therapy in patients with locally advanced hepatocellular carcinoma (DOSISPHERE-01): a randomised, multicentre, open-label phase 2 trial. *Lancet Gastroenterol Hepatol* 2021;6:17–29. [https://doi.org/10.1016/S2468-1253\(20\)30290-9](https://doi.org/10.1016/S2468-1253(20)30290-9).
- [164] Sangro B, Kudo M, Erinjeri JP, Qin S, Ren Z, Chan SL, et al. Durvalumab with or without bevacizumab with transarterial chemoembolisation in hepatocellular carcinoma (EMERALD-1): a multiregional, randomised, double-blind, placebo-controlled, phase 3 study. *Lancet* 2025;405:216–232. [https://doi.org/10.1016/S0140-6736\(24\)02551-0](https://doi.org/10.1016/S0140-6736(24)02551-0).
- [165] **Kudo M, Ren Z, Guo Y, Han G, Lin H, Zheng J, et al.** Transarterial chemoembolization combined with lenvatinib plus pembrolizumab versus dual placebo for unresectable, non-metastatic hepatocellular carcinoma (LEAP-012): a multicentre, randomised, double-blind, phase 3 study. *Lancet* 2025;405:203–215. [https://doi.org/10.1016/S0140-6736\(24\)02575-3](https://doi.org/10.1016/S0140-6736(24)02575-3).
- [166] Dong J, Han G, Ogasawara S, Liu R, Gu S, Liu F, et al. LBA2 TALENTACE: a phase III, open-label, randomized study of on-demand transarterial chemoembolization (TACE) combined with atezolizumab + bevacizumab (Atezo+Bev) or on-demand TACE alone in patients with systemically untreated, intermediate-to-high burden. *Ann Oncol* 2025;36:S62. <https://doi.org/10.1016/j.annonc.2025.05.542>.
- [167] **Peng Z, Fan W, Zhu B, Wang G, Sun J, Xiao C, et al.** Lenvatinib combined with transarterial chemoembolization as first-line treatment for advanced hepatocellular carcinoma: a phase III, randomized clinical trial (LAUNCH). *J Clin Oncol* 2023;41:117–127. <https://doi.org/10.1200/JCO.22.00392>.
- [168] **Kan X-F, Liang B, Zhang X-L, Yu L, Luo Y-C, Zhou S, et al.** Transarterial chemoembolization plus apatinib for unresectable hepatocellular carcinoma: a multicenter, randomized, open-label, phase III trial. *BMC Med* 2025;23:313. <https://doi.org/10.1186/s12916-025-04159-y>.
- [169] **Duan X, Li H, Kuang D, Chen P, Zhang M, Li T, et al.** Comparison of drug-eluting bead transarterial chemoembolization combined with apatinib versus drug-eluting bead transarterial chemoembolization for the treatment of unresectable hepatocellular carcinoma: a randomized, prospective, multicenter phase III trial. *Signal Transduct Target Ther* 2024;9:304. <https://doi.org/10.1038/s41392-024-02012-x>.
- [170] Finn RS, Qin S, Ikeda M, Galle PR, Ducreux M, Kim TY, et al. Atezolizumab plus bevacizumab in unresectable hepatocellular carcinoma. *N Engl J Med* 2020;382:1894–1905. <https://doi.org/10.1056/NEJMoa1915745>.
- [171] **Abou-Alfa GK, Lau G, Kudo M, Chan SL, Kelley RK, Furuse J, et al.** Tremelimumab plus durvalumab in unresectable hepatocellular carcinoma. *NEJM Evid* 2022. <https://doi.org/10.1056/EVIDOA2100070>.
- [172] **Yau T, Galle PR, Decaens T, Sangro B, Qin S, da Fonseca LG, et al.** Nivolumab plus ipilimumab versus lenvatinib or sorafenib as first-line treatment for unresectable hepatocellular carcinoma (CheckMate 9DW): an open-label, randomised, phase 3 trial. *Lancet* 2025;405:1851–1864. [https://doi.org/10.1016/S0140-6736\(25\)00403-9](https://doi.org/10.1016/S0140-6736(25)00403-9).
- [173] **Qin S, Chan SL, Gu S, Bai Y, Ren Z, Lin X, et al.** Camrelizumab plus rivoceranib versus sorafenib as first-line therapy for unresectable hepatocellular carcinoma (CARES-310): a randomised, open-label, international phase 3 study. *Lancet* 2023. [https://doi.org/10.1016/S0140-6736\(23\)00961-3](https://doi.org/10.1016/S0140-6736(23)00961-3).
- [174] Kudo M, Finn RS, Qin S, Han K-H, Ikeda K, Piscaglia F, et al. Lenvatinib versus sorafenib in first-line treatment of patients with unresectable hepatocellular carcinoma: a randomised phase 3 non-inferiority trial. *Lancet* 2018;391:1163–1173. [https://doi.org/10.1016/S0140-6736\(18\)30207-1](https://doi.org/10.1016/S0140-6736(18)30207-1).
- [175] Qin S, Kudo M, Meyer T, Bai Y, Guo Y, Meng Z, et al. Tislelizumab vs sorafenib as first-line treatment for unresectable hepatocellular carcinoma: a phase 3 randomized clinical trial. *JAMA Oncol* 2023. <https://doi.org/10.1001/JAMAONCOL.2023.4003>.
- [176] Llovet JM, Ricci S, Mazzaferro V, Hilgard P, Gane E, Blanc J-F, et al. Sorafenib in advanced hepatocellular carcinoma. *N Engl J Med* 2008;359:378–390. <https://doi.org/10.1056/NEJMoa0708857>.
- [177] Cheng A-L, Kang Y-K, Chen Z, Tsao C-J, Qin S, Kim JS, et al. Efficacy and safety of sorafenib in patients in the Asia-Pacific region with advanced hepatocellular carcinoma: a phase III randomised, double-blind, placebo-controlled trial. *Lancet Oncol* 2009;10:25–34. [https://doi.org/10.1016/S1470-2045\(08\)70285-7](https://doi.org/10.1016/S1470-2045(08)70285-7).
- [178] **Zhou J, Bai L, Luo J, Bai Y, Pan Y, Yang X, et al.** Anlotinib plus penpulimab versus sorafenib in the first-line treatment of unresectable hepatocellular carcinoma (APOLLO): a randomised, controlled, phase 3 trial. *Lancet Oncol* 2025;26:719–731. [https://doi.org/10.1016/S1470-2045\(25\)00190-1](https://doi.org/10.1016/S1470-2045(25)00190-1).
- [179] Lyu N, Wang X, Li J-B, Lai J-F, Chen Q-F, Li S-L, et al. Arterial chemotherapy of oxaliplatin plus fluorouracil versus sorafenib in advanced hepatocellular carcinoma: a biomolecular exploratory, randomized, phase III trial (FOHAIC-1). *J Clin Oncol* 2022;40:468–480. <https://doi.org/10.1200/JCO.21.01963>.
- [180] He M, Li Q, Zou R, Shen J, Fang W, Tan G, et al. Sorafenib plus hepatic arterial infusion of oxaliplatin, fluorouracil, and leucovorin vs sorafenib alone for hepatocellular carcinoma with portal vein invasion. *JAMA Oncol* 2019;5:953. <https://doi.org/10.1001/jamaoncol.2019.0250>.
- [181] Qin S, Bi F, Gu S, Bai Y, Chen Z, Wang Z, et al. Donafenib versus sorafenib in first-line treatment of unresectable or metastatic hepatocellular carcinoma: a randomized, open-label, parallel-controlled phase II-III trial. *J Clin Oncol* 2021;39:3002–3011. <https://doi.org/10.1200/JCO.21.00163>.
- [182] Ren Z, Xu J, Bai Y, Xu A, Cang S, Du C, et al. Sintilimab plus a bevacizumab biosimilar (IBI305) versus sorafenib in unresectable hepatocellular carcinoma (ORIENT-32): a randomised, open-label, phase 2–3 study. *Lancet Oncol* 2021;22:977–990. [https://doi.org/10.1016/S1470-2045\(21\)00252-7](https://doi.org/10.1016/S1470-2045(21)00252-7).
- [183] **Shi Y, Han G, Zhou J, Shi X, Jia W, Cheng Y, et al.** Toripalimab plus bevacizumab versus sorafenib as first-line treatment for advanced hepatocellular carcinoma (HEPATORCH): a randomised, open-label, phase 3 trial. *Lancet Gastroenterol Hepatol* 2025;10:658–670. [https://doi.org/10.1016/S2468-1253\(25\)00059-7](https://doi.org/10.1016/S2468-1253(25)00059-7).
- [184] Vilgrain V, Pereira H, Assenat E, Guib B, Ilonca AD, Pageaux G-P, et al. Efficacy and safety of selective internal radiotherapy with yttrium-90 resin microspheres compared with sorafenib in locally advanced and inoperable hepatocellular carcinoma (SARAH): an open-label randomised controlled phase 3 trial. *Lancet Oncol* 2017;18:1624–1636. [https://doi.org/10.1016/S1470-2045\(17\)30683-6](https://doi.org/10.1016/S1470-2045(17)30683-6).

- [185] Chow PKH, Gandhi M, Tan S-B, Khin MW, Khasbazar A, Ong J, et al. SIRveNIB: selective internal radiation therapy versus sorafenib in Asia-Pacific patients with hepatocellular carcinoma. *J Clin Oncol* 2018;36:1913–1921. <https://doi.org/10.1200/JCO.2017.76.0892>.
- [186] Ricke J, Klumpen HJ, Amthauer H, Bargellini I, Bartenstein P, de Toni EN, et al. Impact of combined selective internal radiation therapy and sorafenib on survival in advanced hepatocellular carcinoma. *J Hepatol* 2019;71(6):1164–1174. <https://doi.org/10.1016/j.jhep.2019.08.006>.
- [187] Fortuny M, De Rosa A, Roca I, Ouchi D, Suárez E, Cadiz A, et al. Hepatocellular carcinoma with macrovascular invasion: review and survival meta-analysis with minimal prognostic criteria. *Hepatology* 2026. <https://doi.org/10.1097/HEP.0000000000001630>.
- [188] Bruix J, Qin S, Merle P, Granito A, Huang Y-H, Bodoky G, et al. Regorafenib for patients with hepatocellular carcinoma who progressed on sorafenib treatment (RESORCE): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 2017;389:56–66. [https://doi.org/10.1016/S0140-6736\(16\)32453-9](https://doi.org/10.1016/S0140-6736(16)32453-9).
- [189] Finn RS, Merle P, Granito A, Huang Y-H, Bodoky G, Pracht M, et al. Outcomes of sequential treatment with sorafenib followed by regorafenib for HCC: additional analyses from the phase III RESORCE trial. *J Hepatol* 2018;69:353–358. <https://doi.org/10.1016/j.jhep.2018.04.010>.
- [190] Abou-Alfa GK, Meyer T, Cheng AL, El-Khoueiry AB, Rimassa L, Ryoo BY, et al. Cabozantinib in patients with advanced and progressing hepatocellular carcinoma. *N Engl J Med* 2018;379:54–63. <https://doi.org/10.1056/NEJMoa1717002>.
- [191] Zhu AX, Kang Y-K, Yen C-J, Finn RS, Galle PR, Llovet JM, et al. Ramucirumab after sorafenib in patients with advanced hepatocellular carcinoma and increased α -fetoprotein concentrations (REACH-2): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Oncol* 2019;20:282–296. [https://doi.org/10.1016/S1470-2045\(18\)30937-9](https://doi.org/10.1016/S1470-2045(18)30937-9).
- [192] Finn RS, Ryoo B-Y, Merle P, Kudo M, Bouattour M, Lim HY, et al. Pembrolizumab as second-line therapy in patients with advanced hepatocellular carcinoma in KEYNOTE-240: a randomized, double-blind, phase III trial. *J Clin Oncol* 2020;38:193–202. <https://doi.org/10.1200/JCO.19.01307>.
- [193] Qin S, Chen Z, Fang W, Ren Z, Xu R, Ryoo B-Y, et al. Pembrolizumab versus placebo as second-line therapy in patients from Asia with advanced hepatocellular carcinoma: a randomized, double-blind, phase III trial. *J Clin Oncol* 2023;41:1434–1443. <https://doi.org/10.1200/JCO.22.00620>.
- [194] Finn RS, Gu K, Chen X, Merle P, Lee K-H, Bouattour M, et al. Second-line pembrolizumab for advanced HCC: meta-analysis of the phase III KEYNOTE-240 and KEYNOTE-394 studies. *JHEP Rep* 2025;7:101350. <https://doi.org/10.1016/j.jhep.2025.101350>.
- [195] Chan SL, Ryoo B-Y, Mo F, Chan LL, Cheon J, Li L, et al. Multicentre phase II trial of cabozantinib in patients with hepatocellular carcinoma after immune checkpoint inhibitor treatment. *J Hepatol* 2024;81:258–264. <https://doi.org/10.1016/j.jhep.2024.03.033>.
- [196] Sanduzzi-Zamparelli M, Martinez SM, Calvo M, Varela M, Llarch N, Iserte G, et al. Interim analysis of the ACTION trial: cabozantinib for hepatocellular carcinoma patients who discontinued first line treatment other than sorafenib or due to sorafenib intolerance. *J Hepatol* 2023;78:S579–S580. [https://doi.org/10.1016/S0168-8278\(23\)01381-8](https://doi.org/10.1016/S0168-8278(23)01381-8).
- [197] Cheon J, Ryoo B-Y, Chon HJ, Kim H-D, Ryu M-H, Kim K-P, et al. Multi-center phase 2 trial of second-line regorafenib in patients with unresectable hepatocellular carcinoma after progression on atezolizumab plus bevacizumab (REGONEXT). *Liver Cancer* 2025;1–21. <https://doi.org/10.1159/000543666>.
- [198] Sanduzzi Zamparelli M, Matilla A, Lledó JL, Martínez SM, Varela M, Iñárraigui M, et al. Early nivolumab addition to regorafenib in patients with hepatocellular carcinoma progressing under first-line therapy (GOING trial), interim analysis and safety profile. *J Clin Oncol* 2022;40:428. https://doi.org/10.1200/JCO.2022.40.4_suppl.428.
- [199] El-Khoueiry AB, Kim T-Y, Blanc J-F, Rosmorduc O, Decaens T, Mathurin P, et al. International, open-label phase 2 study of regorafenib plus pembrolizumab in patients with advanced hepatocellular carcinoma (HCC) previously treated with immune checkpoint inhibitors (ICI). *J Clin Oncol* 2024;42:4007. https://doi.org/10.1200/JCO.2024.42.16_suppl.4007.
- [200] Kim H-D, Sym SJ, Chon HJ, Kim M, Kang JH, Ryoo B-Y, et al. Multicenter phase II trial of lenvatinib in patients with advanced hepatocellular carcinoma after progression on first-line atezolizumab plus bevacizumab. *J Hepatol* 2025. <https://doi.org/10.1016/j.jhep.2025.08.020>.
- [201] Allaire M, Da Fonseca LG, Sanduzzi-Zamparelli M, Choi W-M, Monge C, Liu K, et al. Disparities in access to systemic therapies for patients with hepatocellular carcinoma: an analysis from the International Liver Cancer Association. *Lancet Reg Heal - Eur* 2025;57:101408. <https://doi.org/10.1016/j.lanepe.2025.101408>.
- [202] Iavarone M, Cabibbo G, Biolato M, Della Corte C, Maida M, Barbara M, et al. Predictors of survival in patients with advanced hepatocellular carcinoma who permanently discontinued sorafenib. *Hepatology* 2015;62:784–791. <https://doi.org/10.1002/hep.27729>.
- [203] Ogasawara S, Chiba T, Ooka Y, Suzuki E, Kanogawa N, Saito T, et al. Post-progression survival in patients with advanced hepatocellular carcinoma resistant to sorafenib. *Invest New Drugs* 2016;34:255–260. <https://doi.org/10.1007/s10637-016-0233-1>.
- [204] Reig M, Galle PR, Kudo M, Finn R, Llovet JM, Mett AL, et al. Pattern of progression in advanced hepatocellular carcinoma treated with ramucirumab. *Liver Int* 2021;41:598–607. <https://doi.org/10.1111/liv.14731>.
- [205] Rimassa L, Assenat E, Peck-Radosavljevic M, Pracht M, Zagonel V, Mathurin P, et al. Tivantinib for second-line treatment of MET-high, advanced hepatocellular carcinoma (METIV-HCC): a final analysis of a phase 3, randomised, placebo-controlled study. *Lancet Oncol* 2018;19:682–693. [https://doi.org/10.1016/S1470-2045\(18\)30146-3](https://doi.org/10.1016/S1470-2045(18)30146-3).
- [206] de la Torre-Aláez M, Jordán-Iborra C, Casadei-Gardini A, Bilbao JL, Rodríguez-Fraile M, Sancho L, et al. The pattern of progression defines post-progression survival in patients with hepatocellular carcinoma treated with SIRT. *Cardiovasc Intervent Radiol* 2020;43:1165–1172. <https://doi.org/10.1007/s00270-020-02444-2>.
- [207] Campani C, Vallot A, Ghannouchi H, Allaire M, Evain M, Sultanik P, et al. Impact of radiological response and pattern of progression in patients with HCC treated by atezolizumab-bevacizumab. *Hepatology* 2024;79:49–60. <https://doi.org/10.1097/HEP.0000000000000636>.
- [208] Quinn PL, Saiyed S, Hannon C, Sarna A, Waterman BL, Cloyd JM, et al. Reporting time toxicity in prospective cancer clinical trials: a scoping review. *Support Care Cancer* 2024;32:275. <https://doi.org/10.1007/s00520-024-08487-2>.
- [209] Cheng SJ, Bansal A, Veenstra DL. Productivity loss by cancer stage in patients newly diagnosed with hepatocellular carcinoma: a claims database analysis. *J Manag Care Spec Pharm* 2024;30:572–580. <https://doi.org/10.18553/jmcp.2024.30.6.572>.
- [210] Wörns M-A, Burns D, Paskow M, Makin H, Miller J, Turner LM, et al. Patient experience of hepatocellular carcinoma and their treatment goals: an international qualitative study and patient journey map. *Adv Ther* 2024;41:3598–3614. <https://doi.org/10.1007/s12325-024-02939-1>.
- [211] Lo SH, Sharma R, Costentin CE, Aggio D, Shergill S, Colaone F, et al. Patient preferences for advanced hepatocellular carcinoma treatment: a multicountry stated preference study. *Futur Oncol* 2021;17:4275–4287. <https://doi.org/10.2217/fon-2021-0697>.
- [212] Famularo S, Donadon M, Cipriani F, Giulianti F, Ferri S, Celsa C, et al. Hepatectomy versus sorafenib in advanced nonmetastatic hepatocellular carcinoma. *Ann Surg* 2022;275:743–752. <https://doi.org/10.1097/SLA.0000000000005373>.
- [213] Komatsu S, Tada T, Nishimura T, Tanaka M, Takebe A, Kushida S, et al. Prognosis of hepatectomy versus systemic chemotherapy based on oncological resectability criteria for borderline resectable hepatocellular carcinoma. *Liver Cancer* 2025;1–20. <https://doi.org/10.1159/000546830>.
- [214] Finn RS, Galle PR, Ducreux M, Cheng A-L, Reilly N, Nicholas A, et al. Efficacy and safety of atezolizumab plus bevacizumab versus sorafenib in hepatocellular carcinoma with main trunk and/or contralateral portal vein invasion in IMbrave150. *Liver Cancer* 2024;1–14. <https://doi.org/10.1159/000539897>.
- [215] Matono T, Tada T, Kumada T, Hiraoka A, Hirooka M, Kariyama K, et al. Vascular invasion within the resectability criteria is a prognostic factor in patients treated with atezolizumab and bevacizumab. *Liver Int* 2025;45. <https://doi.org/10.1111/liv.70217>.
- [216] Garin E, Tselikas L, Guiu B, Chalaye J, Rolland Y, de Baere T, et al. Long-term overall survival after selective internal radiation therapy for locally advanced hepatocellular carcinomas: updated analysis of DOSISPHERE-01 trial. *J Nucl Med* 2024;65:264–269. <https://doi.org/10.2967/jnumed.123.266211>.
- [217] Dawson LA, Winter KA, Knox JJ, Zhu AX, Krishnan S, Guha C, et al. Stereotactic body radiotherapy vs sorafenib alone in hepatocellular carcinoma. *JAMA Oncol* 2025;11:136. <https://doi.org/10.1001/jamaoncol.2024.5403>.
- [218] Sangro B, Chan SL, Meyer T, Reig M, El-Khoueiry A, Galle PR. Diagnosis and management of toxicities of immune checkpoint inhibitors in

- hepatocellular carcinoma. *J Hepatol* 2020;72:320–341. <https://doi.org/10.1016/j.jhep.2019.10.021>.
- [219] De Martin E, Fulgenzi CAM, Celsa C, Laurent-Bellue A, Torkpour A, Lombardi P, et al. Immune checkpoint inhibitors and the liver: balancing therapeutic benefit and adverse events. *Gut* 2025;74:1165–1177. <https://doi.org/10.1136/gutjnl-2024-332125>.
- [220] Fortuny M, da Fonseca LG, Allaire M, Sánchez R, Nault J, Iavarone M, et al. Breaking barriers for intensive care admission in patients with advanced HCC on immunotherapy. *Liver Int* 2025;45. <https://doi.org/10.1111/liv.70264>.
- [221] **da Fonseca LG, Piñero F**, Anders M, Bermudez C, Demirdjian E, Varón A, et al. Immune-mediated adverse events following atezolizumab and bevacizumab in a multinational Latin American cohort of unresectable hepatocellular carcinoma. *Oncotarget* 2025;16:348–360. <https://doi.org/10.18632/oncotarget.28721>.
- [222] Campani C, Pallas D, Sidali S, Giouleme O, Blaise L, Grando V, et al. Heterogeneity in adverse events related to atezolizumab-bevacizumab for hepatocellular carcinoma reported in real-world studies. *JHEP Rep* 2024;6:101190. <https://doi.org/10.1016/j.jhep.2024.101190>.
- [223] Fulgenzi CAM, Dalla Pria A, Leone AG, Celsa C, Cabibbo G, Scheiner B, et al. Hepatocellular carcinoma in people living with HIV. *J Hepatol* 2025. <https://doi.org/10.1016/j.jhep.2025.04.028>.
- [224] **El Zarif T, Nassar AH, Adib E**, Fitzgerald BG, Huang J, Mouhieddine TH, et al. Safety and activity of immune checkpoint inhibitors in people living with HIV and cancer: a real-world report from the cancer therapy using checkpoint inhibitors in people living with HIV-international (CATCH-IT) consortium. *J Clin Oncol* 2023;41:3712–3723. <https://doi.org/10.1200/JCO.22.02459>.
- [225] Negri F, Missale G, Antoni AD, Porta C. Hepatocellular cancer therapy in patients with HIV infection: disparities in cancer care, trials enrolment, and cancer-related research. *Transl Oncol* 2021;14:101153. <https://doi.org/10.1016/j.tranon.2021.101153>.
- [226] **Pinato DJ, Allara E**, Chen T-Y, Trevisani F, Minguez B, Zoli M, et al. Influence of HIV infection on the natural history of hepatocellular carcinoma: results from a global multicohort study. *J Clin Oncol* 2019;37:296–304. <https://doi.org/10.1200/JCO.18.00885>.
- [227] **Magyar CTJ, O’Kane GM**, Aceituno L, Li Z, Vogel A, Bruix J, et al. Liver transplantation for hepatocellular carcinoma: an expanding cornerstone of care in the era of immunotherapy. *J Clin Oncol* 2025;43:589–604. <https://doi.org/10.1200/JCO.24.00857>.
- [228] Marino R, Hassan AT, Fagenson A, Tabrizian P. Liver transplantation for hepatocellular carcinoma following immunotherapy. *Curr Opin Organ Transpl* 2025;30:242–250. <https://doi.org/10.1097/MOT.0000000000001228>.
- [229] Ye Z, Song M, Ama A-K, Zhuang L, Zhang Y, Ye S, et al. Prognostic impact of tace combined with sorafenib or lenvatinib followed by regorafenib in recurrent hepatocellular carcinoma after liver transplantation: a retrospective study. *BMC Cancer* 2025;25:1049. <https://doi.org/10.1186/s12885-025-14446-9>.
- [230] Iavarone M, Invernizzi F, Ivanics T, Mazza S, Zavaglia C, Sanduzzi-Zamparelli M, et al. Regorafenib efficacy after sorafenib in patients with recurrent hepatocellular carcinoma after liver transplantation: a retrospective study. *Liver Transpl* 2021;27:1767–1778. <https://doi.org/10.1002/lt.26264>.
- [231] Iavarone M, Invernizzi F, Czuderna C, Sanduzzi-Zamparelli M, Bhoori S, Amadeo G, et al. Preliminary experience on safety of regorafenib after sorafenib failure in recurrent hepatocellular carcinoma after liver transplantation. *Am J Transpl* 2019;19:3176–3184. <https://doi.org/10.1111/ajt.15551>.
- [232] Sakaguchi K, Satoh T, Kawaguchi S, Aoyama T, Asahara K, Endo S, et al. Safety and efficacy of atezolizumab plus bevacizumab combination therapy in patients with unresectable hepatocellular carcinoma undergoing dialysis: a case series. *Clin J Gastroenterol* 2024;17:515–522. <https://doi.org/10.1007/s12328-024-01946-7>.
- [233] **Díaz-González Á, Sanduzzi-Zamparelli M, da Fonseca LG**, Di Costanzo GG, Alves R, Iavarone M, et al. International and multicenter real-world study of sorafenib-treated patients with hepatocellular carcinoma under dialysis. *Liver Int* 2020;40:1467–1476. <https://doi.org/10.1111/liv.14436>.
- [234] Fortuny M, Sanduzzi-Zamparelli M, Reig M. Systemic therapies in hepatocellular carcinoma: a revolution? *United Eur Gastroenterol J* 2024;12:252–260. <https://doi.org/10.1002/ueg2.12510>.
- [235] Eisenhauer EA, Therasse P, Bogaerts J, Schwartz LH, Sargent D, Ford R, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer* 2009;45:228–247. <https://doi.org/10.1016/j.ejca.2008.10.026>.
- [236] Bruix J, Sherman M, Llovet JM, Beaugrand M, Lencioni R, Burroughs AK, et al. Clinical management of hepatocellular carcinoma. Conclusions of the barcelona-2000 EASL conference. *J Hepatol* 2001;35:421–430. [https://doi.org/10.1016/S0168-8278\(01\)00130-1](https://doi.org/10.1016/S0168-8278(01)00130-1).
- [237] Kudo M, Ueshima K, Kubo S, Sakamoto M, Tanaka M, Ikai I, et al. Response evaluation criteria in cancer of the liver (RECICL) (2015 revised version). *Hepatol Res* 2016;46:3–9. <https://doi.org/10.1111/hepr.12542>.
- [238] Lencioni R, Llovet J. Modified RECIST (mRECIST) assessment for hepatocellular carcinoma. *Semin Liver Dis* 2010;30:52–60. <https://doi.org/10.1055/s-0030-1247132>.
- [239] Choi H, Chamsangavej C, Faria SC, Macapinlac HA, Burgess MA, Patel SR, et al. Correlation of computed tomography and positron emission tomography in patients with metastatic gastrointestinal stromal tumor treated at a single institution with imatinib mesylate: proposal of new computed tomography response criteria. *J Clin Oncol* 2007;25:1753–1759. <https://doi.org/10.1200/JCO.2006.07.3049>.
- [240] Seymour L, Bogaerts J, Perrone A, Ford R, Schwartz LH, Mandrekas S, et al. iRECIST: guidelines for response criteria for use in trials testing immunotherapeutics. *Lancet Oncol* 2017;18:e143–e152. [https://doi.org/10.1016/S1470-2045\(17\)30074-8](https://doi.org/10.1016/S1470-2045(17)30074-8).
- [241] Hodi FS, Ballinger M, Lyons B, Soria J-C, Nishino M, Tabernero J, et al. Immune-modified response evaluation criteria in solid tumors (imRECIST): refining guidelines to assess the clinical benefit of cancer immunotherapy. *J Clin Oncol* 2018;36:850–858. <https://doi.org/10.1200/JCO.2017.75.1644>.
- [242] <https://www.acr.org/Clinical-Resources/Clinical-Tools-and-Reference/Reporting-and-Data-Systems/LI-RADS> n.d.
- [243] Yao FY, Ferrell L, Bass NM, Watson JJ, Bacchetti P, Venook A, et al. Liver transplantation for hepatocellular carcinoma: expansion of the tumor size limits does not adversely impact survival. *Hepatology* 2001;33:1394–1403. <https://doi.org/10.1053/jhep.2001.24563>.
- [244] Xu E, Tabrizian P, Gutierrez J, Hoteit M, Ghaziani T, Zhou K, et al. Downstaging of hepatocellular carcinoma before liver transplantation: Results from a national multicenter prospective cohort study. *Hepatology* 2025;82(3):612–625. <https://doi.org/10.1097/HEP.0000000000001231>. Epub 2025 Jan 14.
- [245] Tran NH, Muñoz S, Thompson S, Hallemeier CL, Bruix J. Hepatocellular carcinoma downstaging for liver transplantation in the era of systemic combined therapy with anti-VEGF/TKI and immunotherapy. *Hepatology* 2022;76:1203–1218. <https://doi.org/10.1002/hep.32613>.
- [246] **Guo Z, Liu Y, Ling Q, Xu L, Wang T, Zhu J**, et al. Pretransplant use of immune checkpoint inhibitors for hepatocellular carcinoma: a multicenter, retrospective cohort study. *Am J Transpl* 2024;24:1837–1856. <https://doi.org/10.1016/j.ajt.2024.04.007>.
- [247] Tabrizian P, Holzner ML, Ajmera V, Kim AK, Zhou K, Schnickel GT, et al. Intention-to-treat outcomes of patients with hepatocellular carcinoma receiving immunotherapy before liver transplant: the multicenter VITALITY study. *J Hepatol* 2025;82:512–522. <https://doi.org/10.1016/j.jhep.2024.09.003>.
- [248] Rezaee-Zavareh MS, Yeo YH, Wang T, Guo Z, Tabrizian P, Ward SC, et al. Impact of pre-transplant immune checkpoint inhibitor use on post-transplant outcomes in HCC: a systematic review and individual patient data meta-analysis. *J Hepatol* 2025;82:107–119. <https://doi.org/10.1016/j.jhep.2024.06.042>.
- [249] **Yalikun K, Li Z, Zhang J**, Chang Z, Li M, Sun Z, et al. Hepatic artery infusion chemotherapy combined with camrelizumab and apatinib as conversion therapy for patients with unresectable hepatocellular carcinoma: a single-arm exploratory trial. *BMC Cancer* 2025;25:838. <https://doi.org/10.1186/s12885-025-14250-5>.
- [250] **Li L, Xu X, Wang W**, Huang P, Yu L, Ren Z, et al. Safety and efficacy of PD-1 inhibitor (sintilimab) combined with transarterial chemoembolization as the initial treatment in patients with intermediate-stage hepatocellular carcinoma beyond up-to-seven criteria. *J Immunother Cancer* 2025;13:e010035. <https://doi.org/10.1136/jitc-2024-010035>.
- [251] **Chen Y, Zhang J, Hu W**, Li X, Sun K, Shen Y, et al. Envafoimab plus lenvatinib and transcatheter arterial chemoembolization for unresectable hepatocellular carcinoma: a prospective, single-arm, phase II study. *Signal Transduct Target Ther* 2024;9:280. <https://doi.org/10.1038/s41392-024-01991-1>.
- [252] Liu D, Mu H, Liu C, Zhang W, Cui Y, Wu Q, et al. Sintilimab, bevacizumab biosimilar, and HAIC for unresectable hepatocellular carcinoma conversion therapy: a prospective, single-arm phase II trial. *Neoplasma* 2024;70:811–818. https://doi.org/10.4149/neo_2023_230806N413.

- [253] **Zhang W, Tong S, Hu B**, Wan T, Tang H, Zhao F, et al. Lenvatinib plus anti-PD-1 antibodies as conversion therapy for patients with unresectable intermediate-advanced hepatocellular carcinoma: a single-arm, phase II trial. *J Immunother Cancer* 2023;11:e007366. <https://doi.org/10.1136/jitc-2023-007366>.
- [254] Goto Y, Niizeki T, Fukutomi S, Shirono T, Shimose S, Iwamoto H, et al. Preoperative chemotherapy followed by hepatectomy for potentially resectable UICC7 stage IIIA, IIIB hepatocellular carcinoma; A phase II clinical trial. *Kurume Med J* 2021;68:MS6834010. <https://doi.org/10.2739/kurumemedj.MS6834010>.
- [255] Chiang CL, Chiu KWH, Chan KSK, Lee FAS, Li JCB, Wan CWS, et al. Sequential transarterial chemoembolisation and stereotactic body radiotherapy followed by immunotherapy as conversion therapy for patients with locally advanced, unresectable hepatocellular carcinoma (START-FIT): a single-arm, phase 2 trial. *Lancet Gastroenterol Hepatol* 2023;8:169–178. [https://doi.org/10.1016/S2468-1253\(22\)00339-9](https://doi.org/10.1016/S2468-1253(22)00339-9).
- [256] **Shen L, Cao F, Liu Y**, Nuerhashi G, Lin L, Tan H, et al. Hepatic artery infusion of FOLFOX chemotherapy plus camrelizumab combined with sorafenib for advanced hepatocellular carcinoma in Barcelona Clinic Liver Cancer stage C (Double-IA-001): a phase II trial. *BMC Med* 2025;23:275. <https://doi.org/10.1186/s12916-025-04110-1>.
- [257] González-Sánchez H, Castaño-García A, Celada-Sendino M, Flórez-Díez P, García-Calonge M, Rodríguez M, et al. Epidemiological characteristics of contemporary patients with untreated hepatocellular carcinoma. *Rev Española Enfermedades Dig* 2025. <https://doi.org/10.17235/reed.2025.11029/2024>.
- [258] D'Amico G, Bernardi M, Angeli P. Towards a new definition of decompensated cirrhosis. *J Hepatol* 2022;76:202–207. <https://doi.org/10.1016/j.jhep.2021.06.018>.
- [259] Premkumar M, Kamath PS, Reiberger T, Reddy KR. Recompensation in decompensated cirrhosis. *Lancet Gastroenterol Hepatol* 2025. [https://doi.org/10.1016/S2468-1253\(25\)00095-0](https://doi.org/10.1016/S2468-1253(25)00095-0).
- [260] Piano S, Reiberger T, Bosch J. Mechanisms and implications of recompensation in cirrhosis. *JHEP Rep* 2024;6:101233. <https://doi.org/10.1016/j.jhepr.2024.101233>.
- [261] Hofer BS, Simbrunner B, Hartl L, Jachs M, Balcar L, Paternostro R, et al. Hepatic recompensation according to Baveno VII criteria is linked to a significant survival benefit in decompensated alcohol-related cirrhosis. *Liver Int* 2023;43:2220–2231. <https://doi.org/10.1111/liv.15676>.
- [262] Hofer BS, Simbrunner B, Hartl L, Jachs M, Bauer DJM, Balcar L, et al. Alcohol abstinence improves prognosis across all stages of portal hypertension in alcohol-related cirrhosis. *Clin Gastroenterol Hepatol* 2023;21:2308–2317.e7. <https://doi.org/10.1016/j.cgh.2022.11.033>.
- [263] Marcellin P, Gane E, Buti M, Afdhal N, Sievert W, Jacobson IM, et al. Regression of cirrhosis during treatment with tenofovir disoproxil fumarate for chronic hepatitis B: a 5-year open-label follow-up study. *Lancet* 2013;381:468–475. [https://doi.org/10.1016/S0140-6736\(12\)61425-1](https://doi.org/10.1016/S0140-6736(12)61425-1).
- [264] **Semmler G, Lens S**, Hidalgo Á, Alonso López S, Perez-Perez M, Dajti E, et al. Incidence and clinical significance of recompensation after HCV cure. *Clin Gastroenterol Hepatol* 2025. <https://doi.org/10.1016/j.cgh.2025.04.026>.
- [265] Sánchez-Torrijos Y, Fernández-Álvarez P, Rosales JM, Pérez-Estrada C, Alañón-Martínez P, Rodríguez-Perálvarez M, et al. Prognostic implications of recompensation of decompensated cirrhosis after SVR in patients with hepatitis C. *J Hepatol* 2025. <https://doi.org/10.1016/j.jhep.2025.02.041>.
- [266] Soliman H, Ringash J, Jiang H, Singh K, Kim J, Dinniwell R, et al. Phase II trial of palliative radiotherapy for hepatocellular carcinoma and liver metastases. *J Clin Oncol* 2013;31:3980–3986. <https://doi.org/10.1200/JCO.2013.49.9202>.
- [267] Iserte G, Llarch N, Moon AM, Reig M. Optimizing liver cancer care through BCLC principles. *Semin Liver Diseases*. <https://doi.org/10.1055/a-2769-7224>.

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